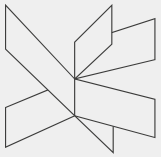
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ELFORSK

QThermonet

A thermonet dimensioning tool for QGIS

Tutorial 1 (v. 0.2.3 – October 2026)

Data Licenses

Please attribute QThermonet/pythermonet with a link to:

- Erbs Poulsen, S., & Tordrup, K. (2025). An integrated design model for ambient temperature district heating and cooling networks. Science and Technology for the Built Environment, 1-10. https://doi.org/10.1080/23744731.2025.2523198
- https://github.com/BYKVAD/pythermonet/tree/legacy
- plugins.qgis.org/plugins/qthermonet_processing_v_0_1/
- https://www.ucviden.dk/da/publications/qthermonet-a-thermonet-dimensioning-tool-for-qgis/

Data used in this tutorial is licensed under an Open Database License

Introduction/1

- This tutorial helps new users get started with QThermonet – a QGIS-based tool for fast and simple thermonet dimensioning.

- QThermonet integrates the free and open-source dimensioning tool pythermonet (Erbs Poulsen & Tordrup, 2025) as a plugin within the desktop QGIS.

- To complete this tutorial basic knowledge of QGIS is required.

Erbs Poulsen, S., & Tordrup, K. (2025). An integrated design model for ambient temperature district heating and cooling networks. Science and Technology for the Built Environment, 1-10. https://doi.org/10.1080/23744731.2025.2523198

Introduction/2

Before getting started the following QGIS software must be downloaded and installed.

- QGIS version 3.44.15

This tutorial was successfully performed within Windows 11 OS and using QGIS version 3.44.15

Not compatible with QGIS 4 – use QThermonet v1.0 or later for QGIS 4

Introduction/3

This tutorial is divided in two parts.

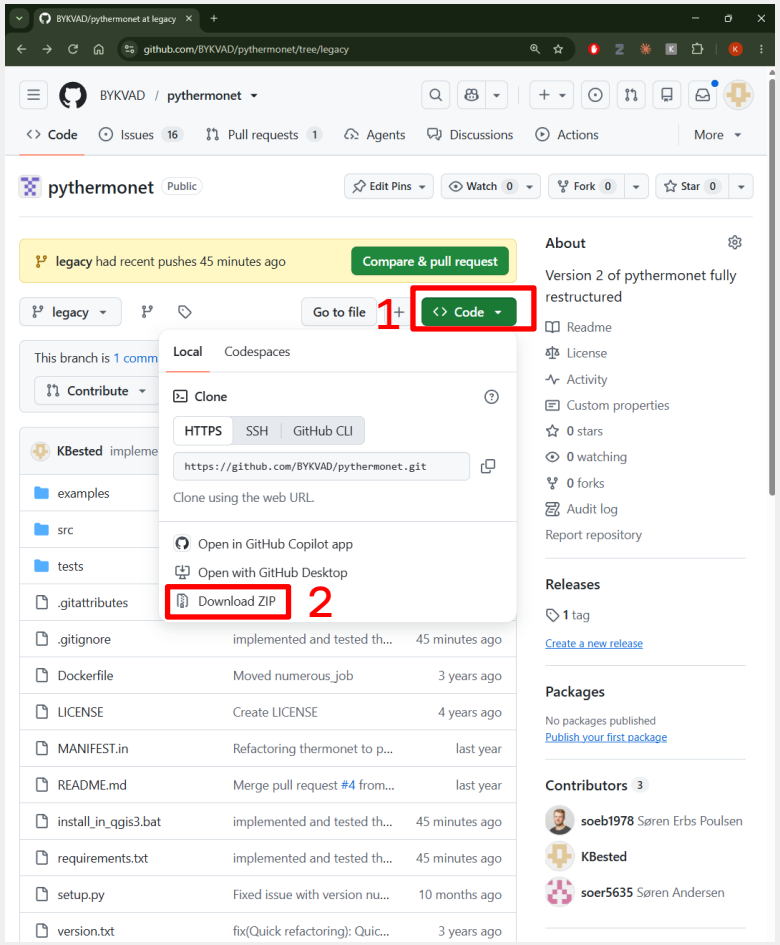
Part A covers the installation of pythermonet and QThermonet, and required access to Datafordeler.dk

Part B provides a step-by-step example of how to dimension a simple thermonet using real data from a Danish case. This is expected to take **ca. 1 hour.**

Part A / Installing pythermonet 1

Please visit https://github.com/BYKVAD/pythermonet/tree/legacy to download a copy of the legacy version of pythermonet (Erbs Poulsen & Tordrup 2025)

Unzip the downloaded folder to your desktop or C-drive

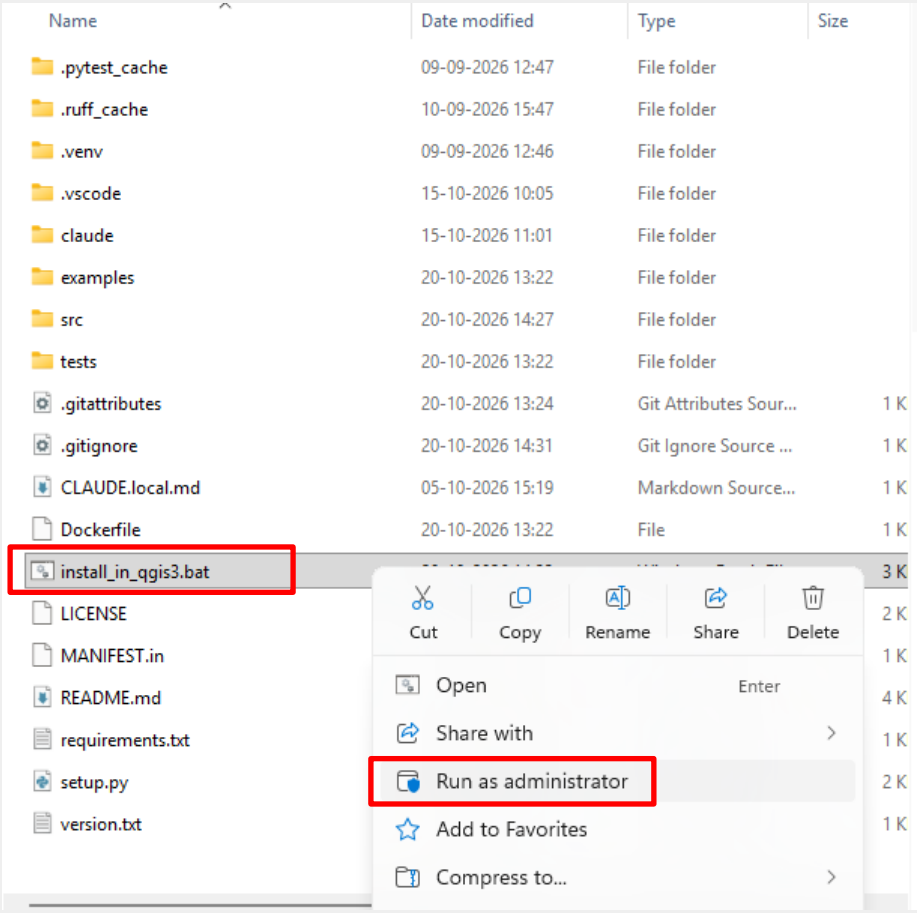


*Erbs Poulsen, S., & Tordrup, K. (2025). An integrated design model for ambient temperature district heating and cooling networks. Science and Technology for the Built Environment, 1-10. https://doi.org/10.1080/23744731.2025.2523198

Part A / Installing pythermonet 2

Next, you need to make a few changes to connect the pythermonet codes with QGIS so that the plugin will work.

1. Run install_in_qgis3.bat as an administrator (right-click and choose run as administrator)
2. A window will pop up and try to install pythermonet. If it works, you will see a message saying it was successful.
3. You can click any key to continue and the window will close.

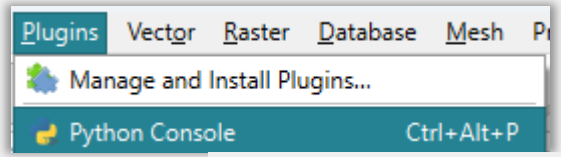


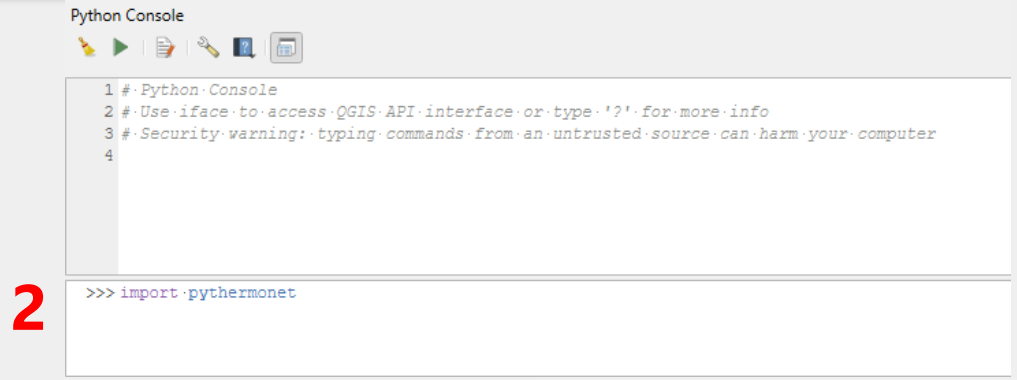
Part A / Installing pythermonet 3

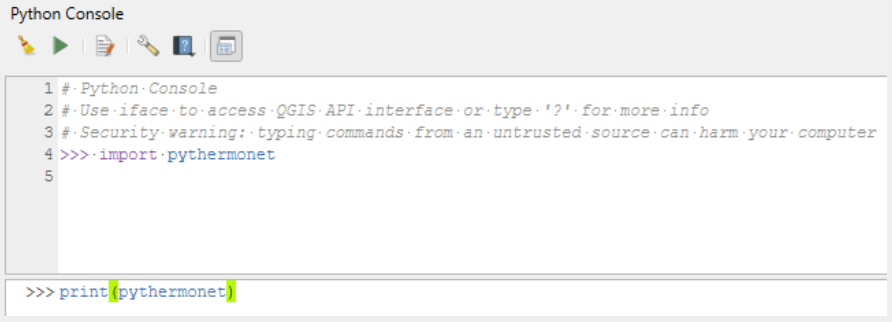
Check that the installation was successful by

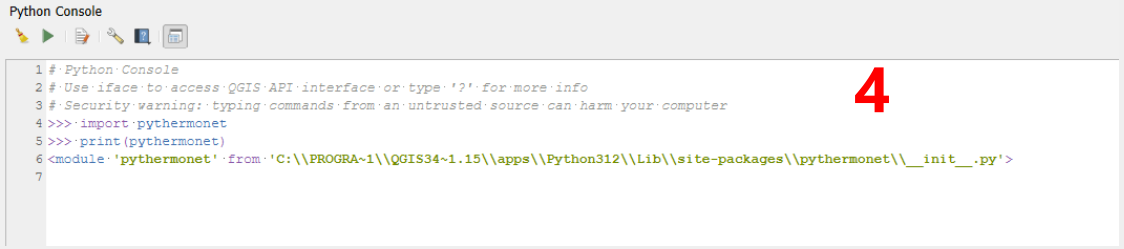
- Restarting QGIS
- Opening the QGIS "Python Console"
- Run 'import pythermonet'
- Run 'print(pythermonet)'
- Check that the path is correct

NB: if you are re-installing pythermonet, you need to delete the old version or change the filepath.



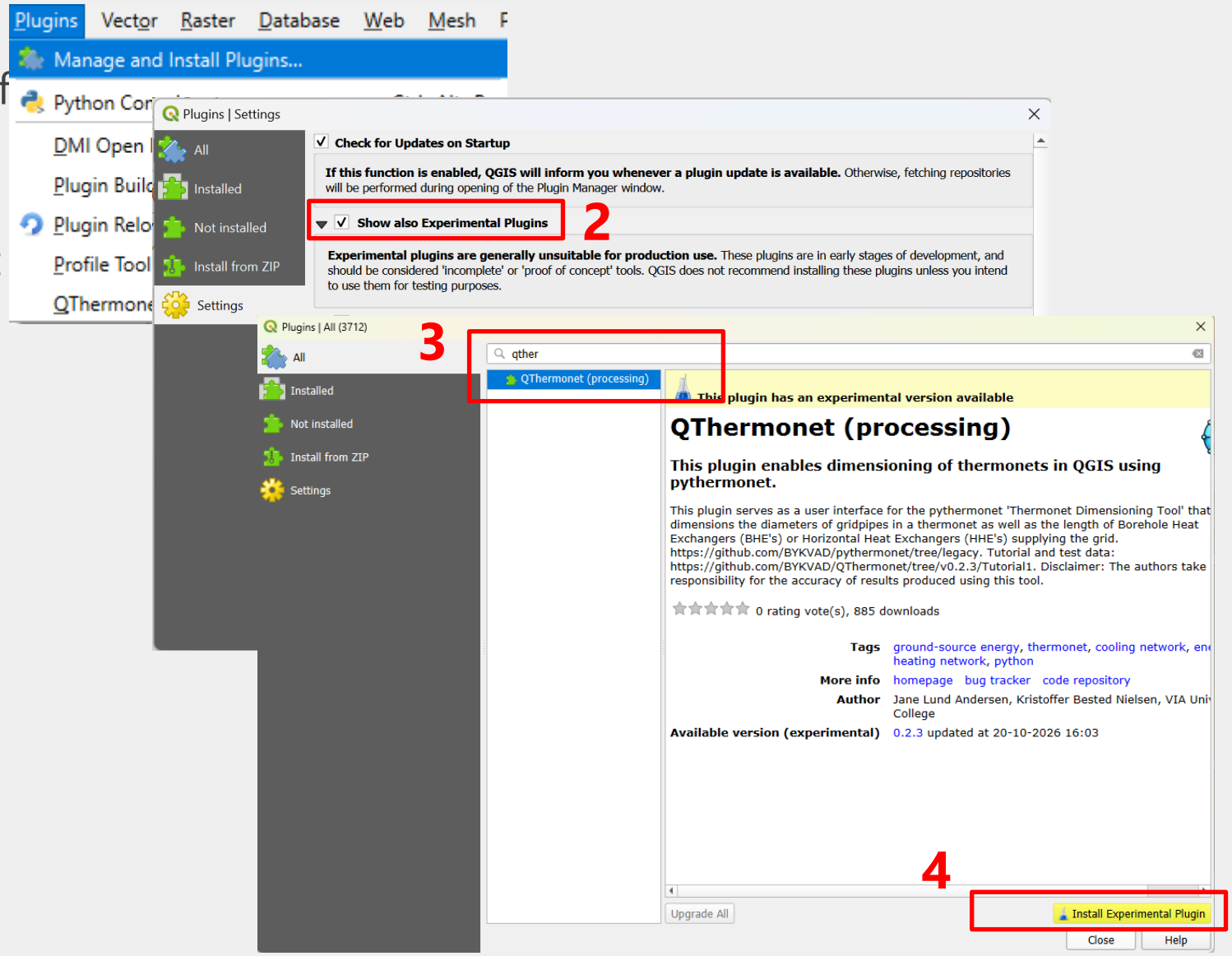






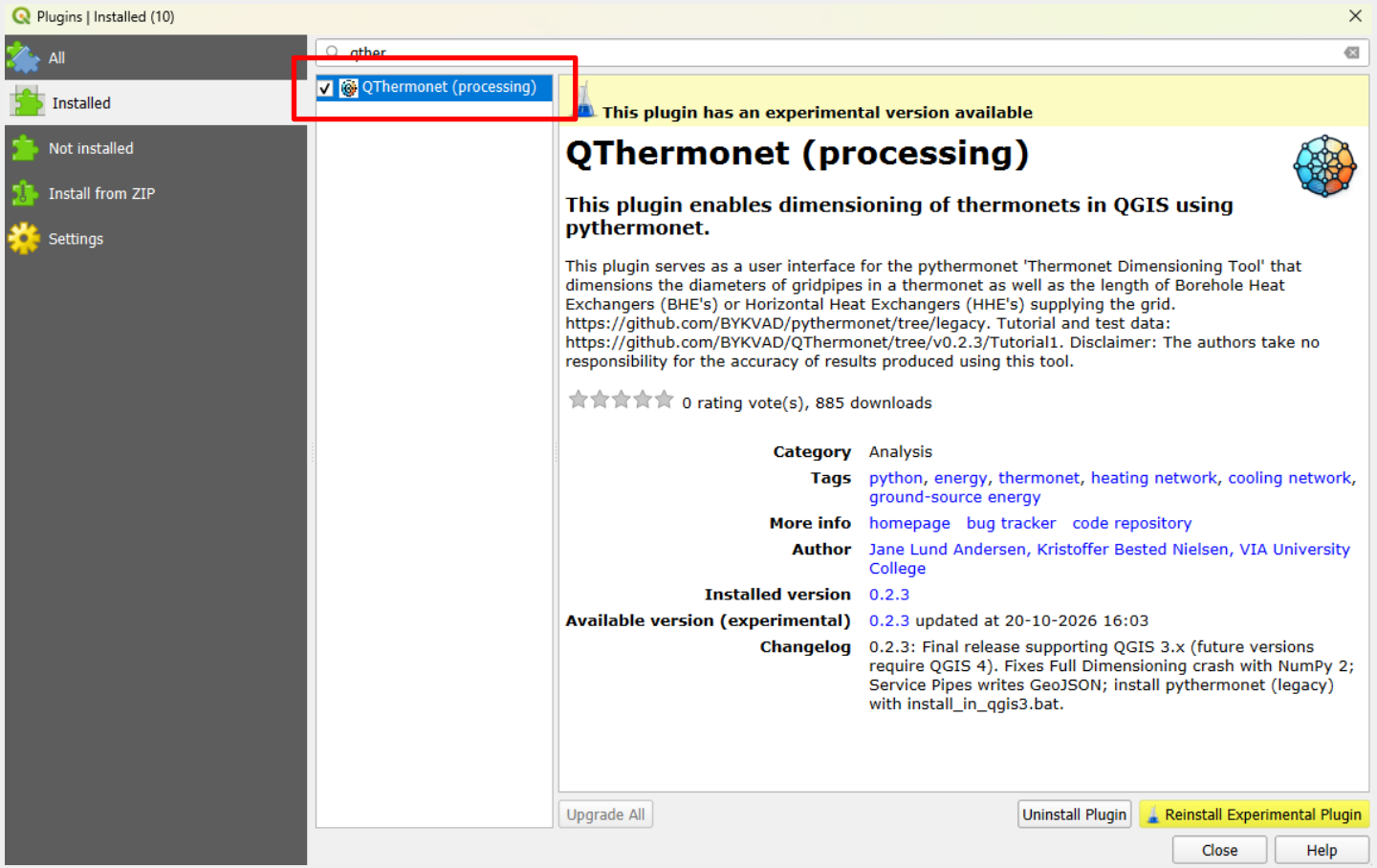
Part A / Installing QThermonet plugin 1

- First, ensure you have a recent installation of QGIS: 3.44 or later (performance tested on 3.44.15)
- To get the latest version of the QThermonet plugin for QGIS go to **Plugins** in the QGIS menu bar and click on 'Manage and Install Plugins…'
- In the dialogue window that opens go to **Settings** and check **Show also Experimental Plugins**
- Go to **All** and search for **QThermonet**
- Click **Install Experimental Plugin**



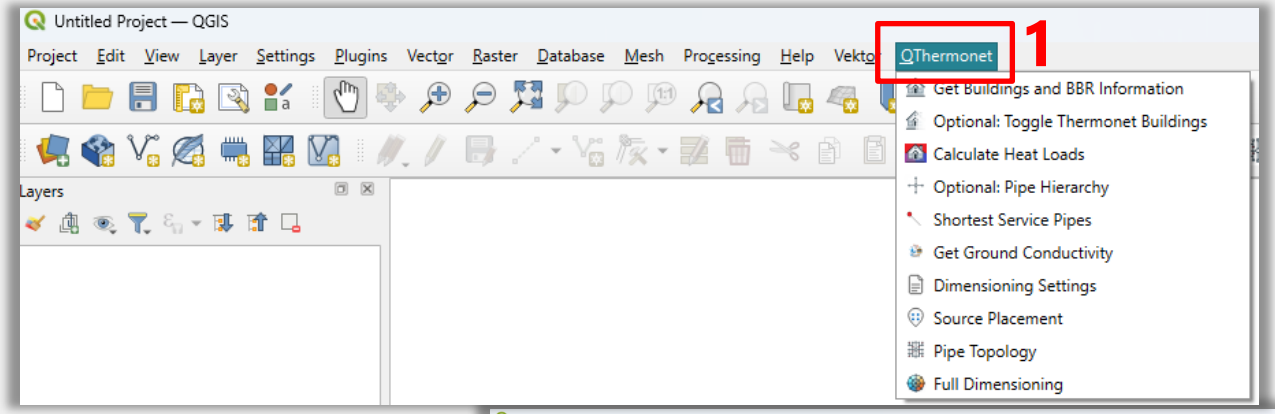
Part A / Installing QThermonet plugin 2

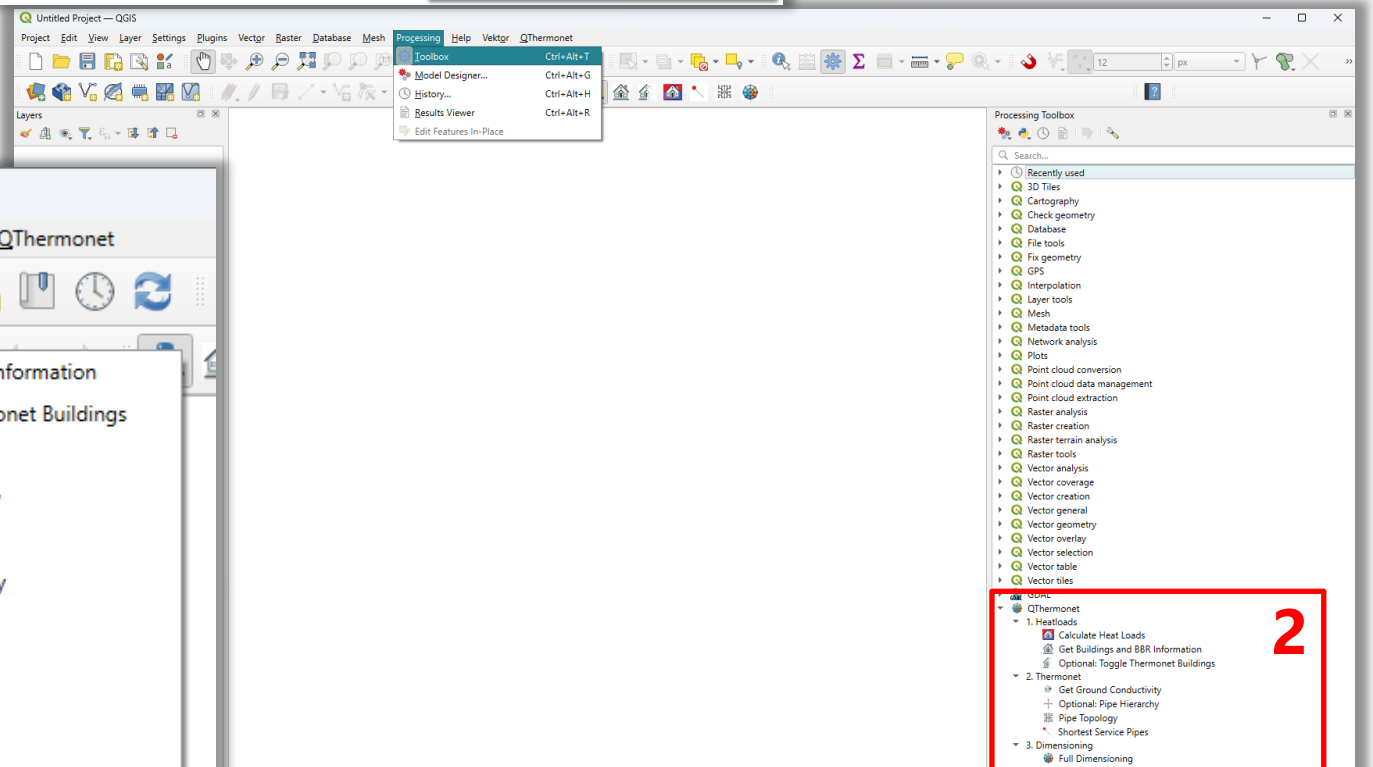
- QGIS will let you know whether the installation was successful.
- Make sure the plugin is checked and close the dialogue window.

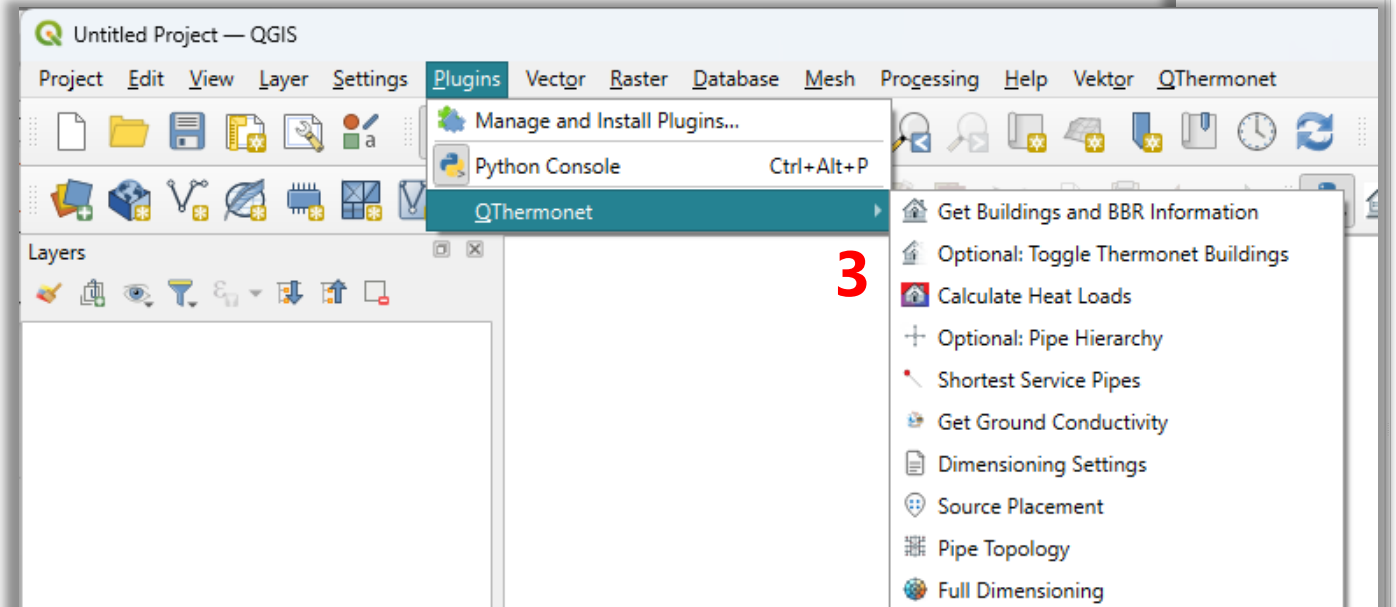


Part A / Installing QThermonet plugin 3

- The **QThermonet** plugin should now appear in three places:
  1. The menu-bar
  2. The processing toolbox
  3. In a submenu to the Plugins menu.



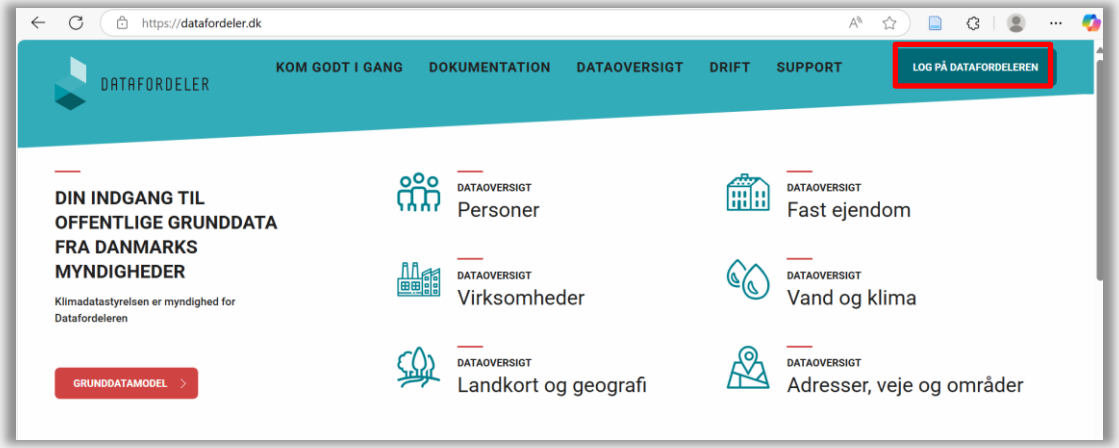




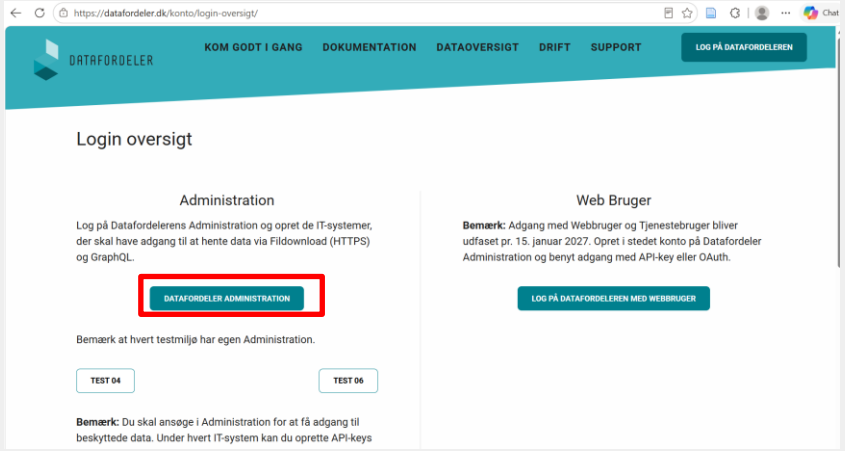
Part A / Datafordeler access 1

To calculate heat loads for existing buildings in Denmark you need to retrieve BBR information within the QThermonet plugin.

For this, you need a login to Datafordeler.dk which you can get after creating a user (ctrl+click left image to follow link):



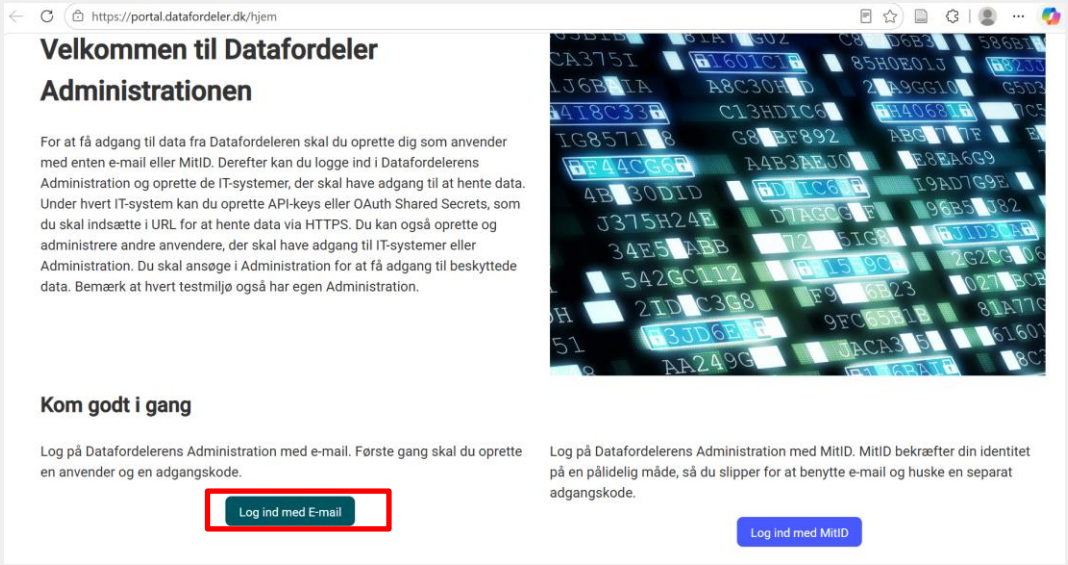
Select 'Administration' in the next step

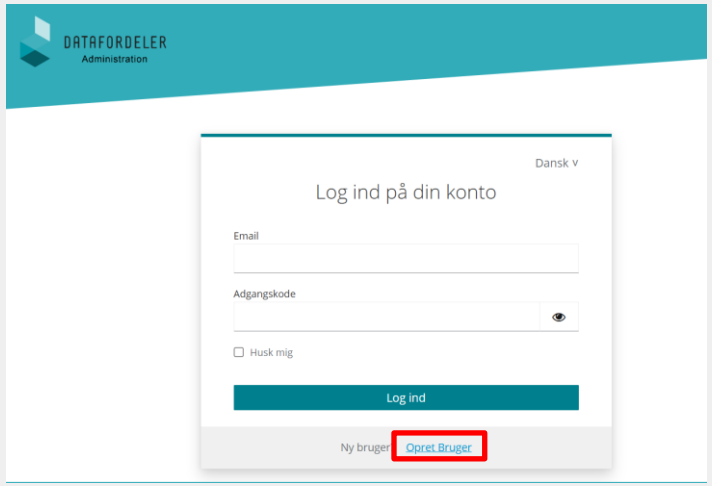


Part A / Datafordeler access 2

If you have not used Datafordeler.dk before, you need to create a new user ('Log ind med e-mail' then 'Opret bruger' - follow the guidelines on the homepage).

Else you can log on with your existing user and then NemLog-in + MitID



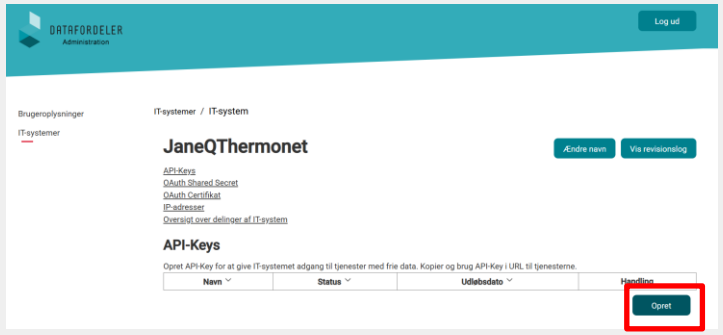


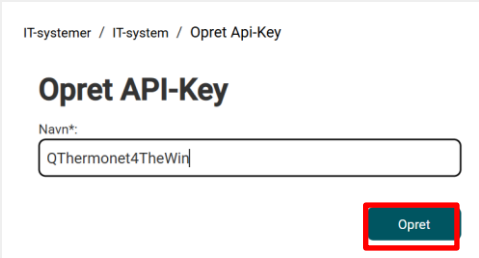
Part A / Datafordeler access 3

After entering the system you have to:

- Click ‘Opret’ below IT-systemer and choose a name
- In the section API-keys select ‘opret’ and choose a name for your API key
- Copy the API key to somewhere on your computer



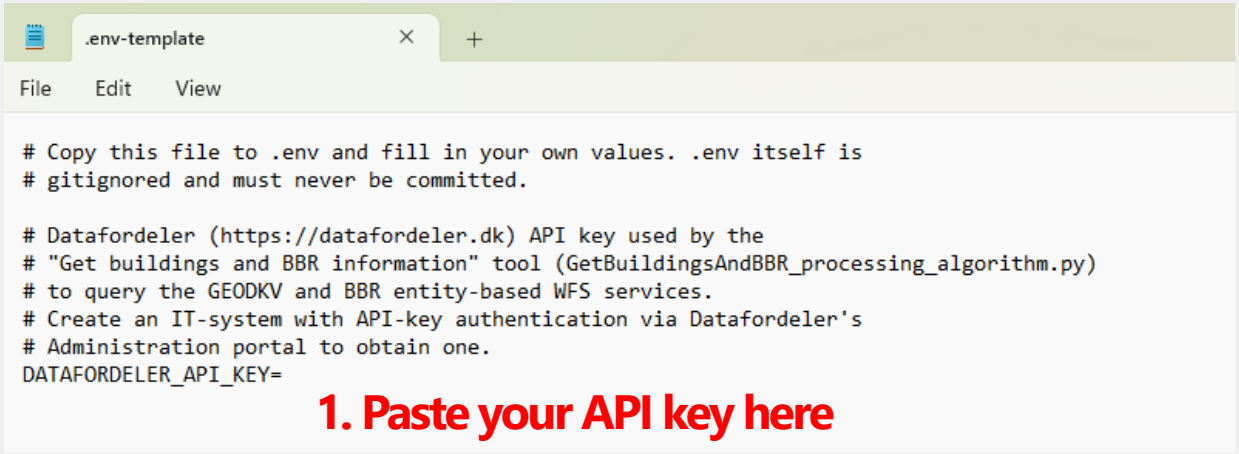


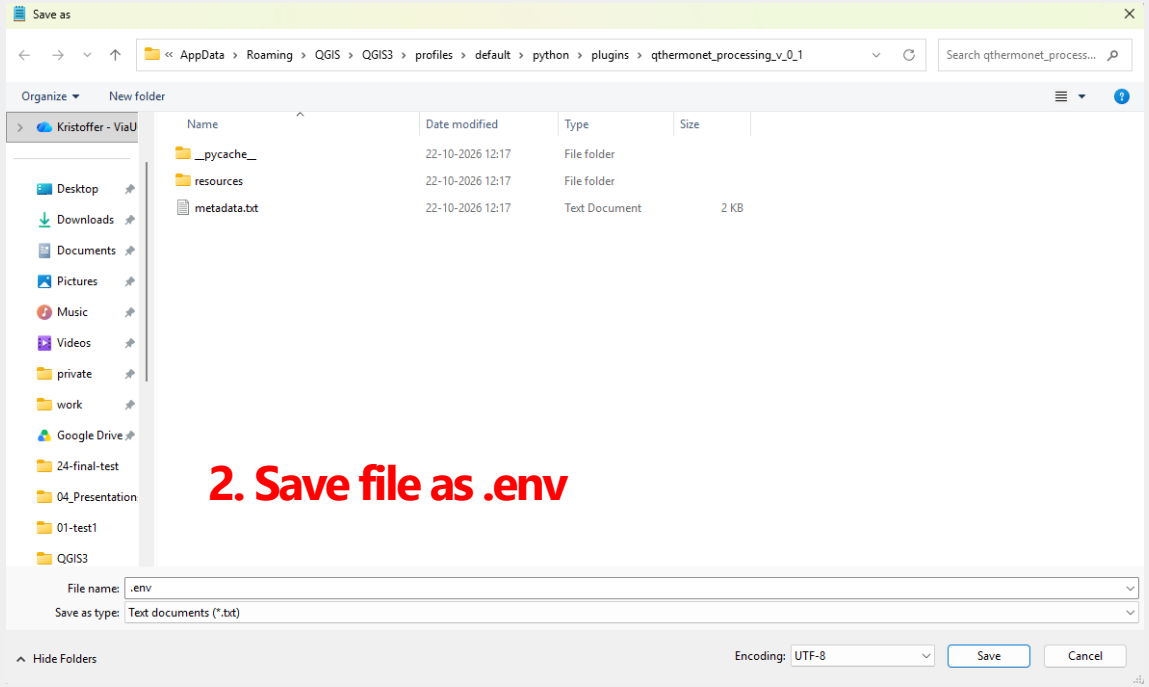


Part A / Datafordeler access 4

After retrieving your API key from Datafordeler, you have to enter it into the .env-template in your QThermonet folder (open it in a text-editor).

"%APPDATA%\QGIS\QGIS3\profiles\default\python\plugins\qthermonet_processing_v_0_1\"
Then save the file with the API key as .env





Part B / Test Case Overview

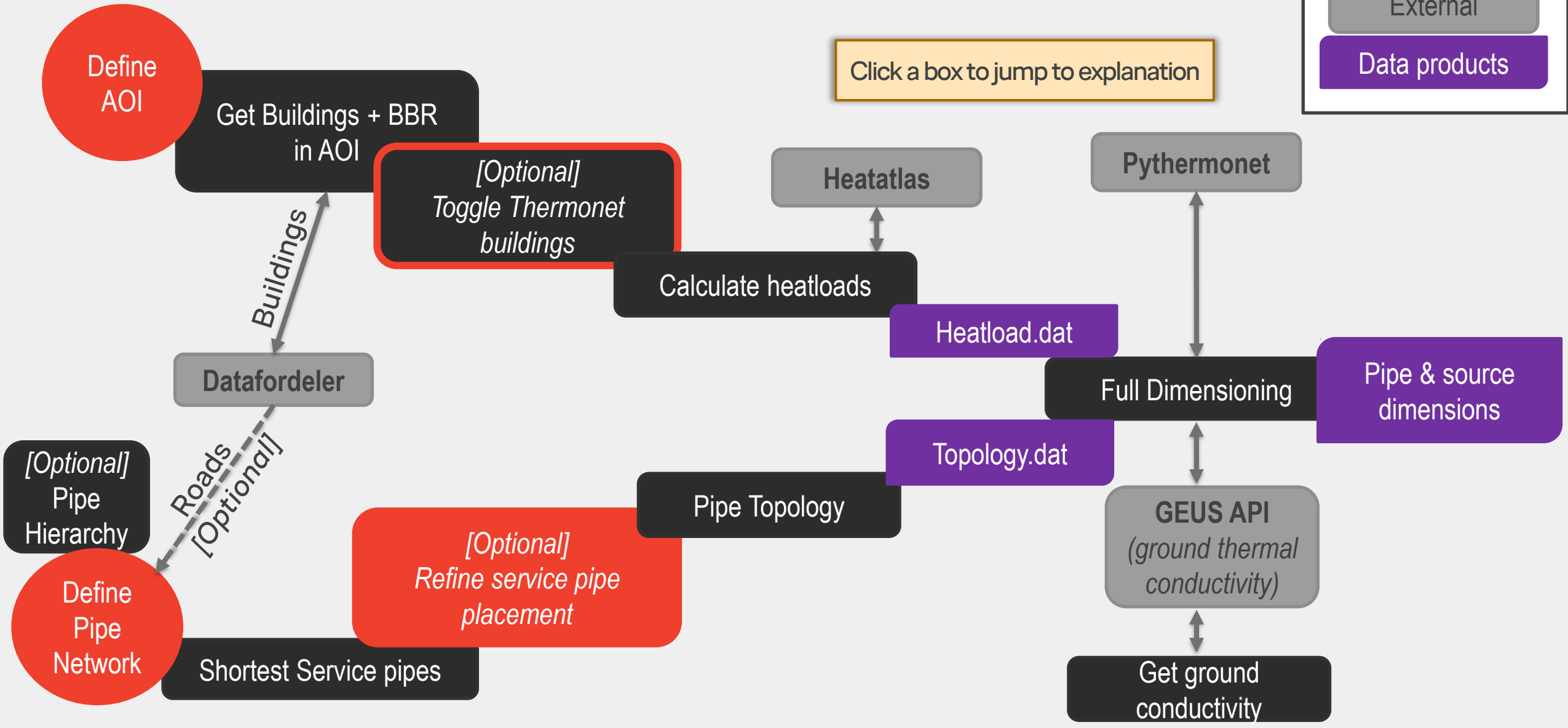
Part B guides the user through a complete thermonet dimensioning example using a Danish test case

It is divided in three parts:

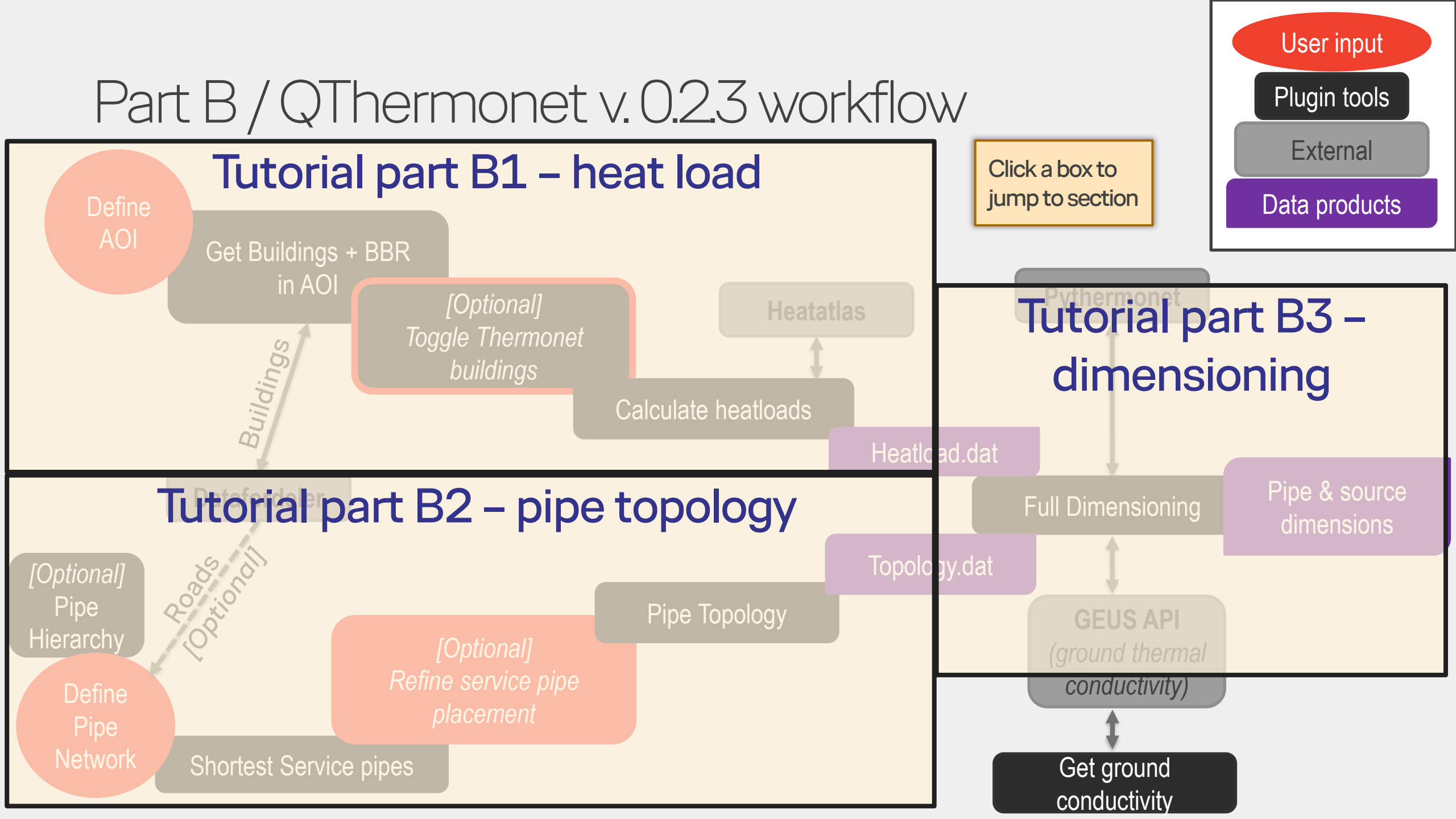
- In part **B1,** the expected heat loads are calculated for a defined Area-of-interest (AOI)
- In part **B2**, the thermonet pipe layout and topology is derived
- Finally, in part **B3**, the dimensioning of the thermonet source is performed

The workflow diagram on the following slides shows how the different tools are connected

Part B / QThermonet v. 0.2.3 workflow



Part B / QThermonet v. 0.2.3 workflow

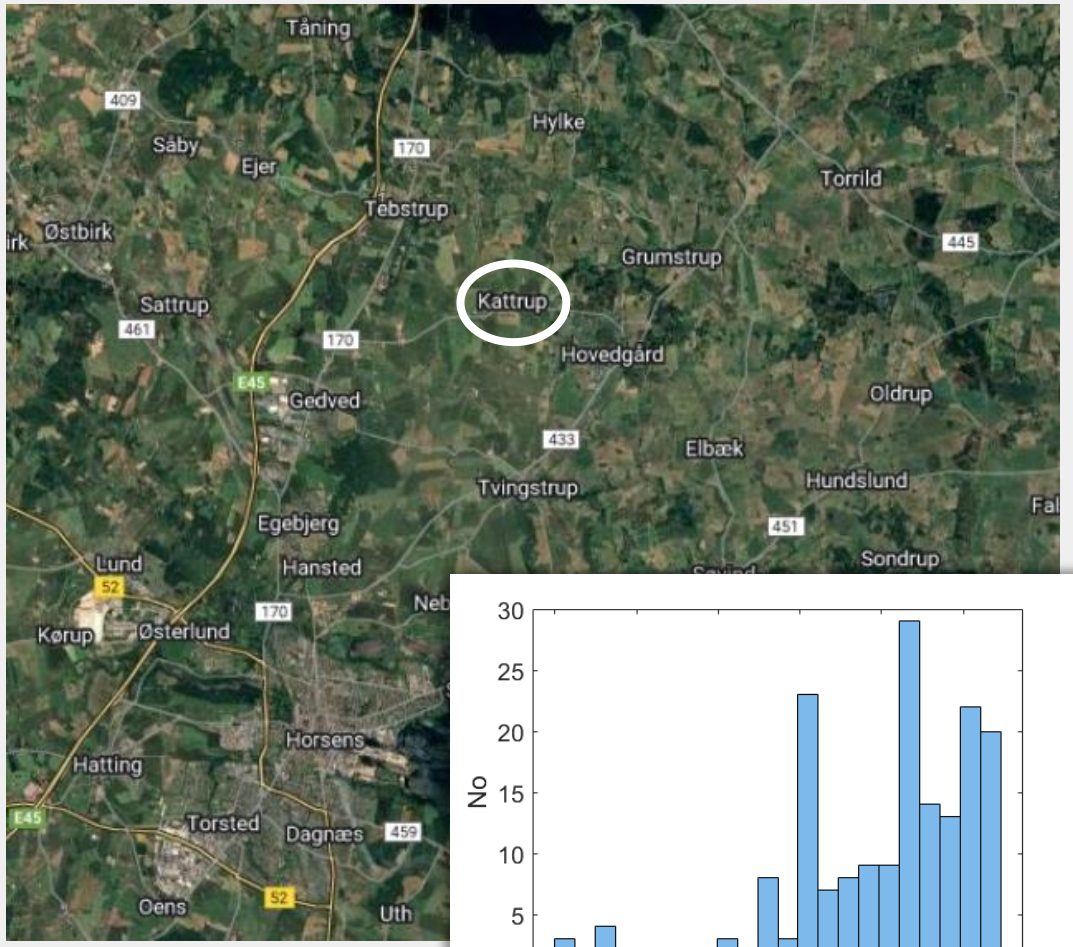


Part B / Test case setting

The test case area that we will work with in this tutorial is located in the village of Kattrup, Horsens Kommune, Denmark. In Kattrup, most buildings are currently heated by natural gas or oil. Buildings are from 1751-2021.

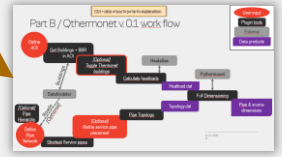
According to the climate plan for Horsens Kommune, Kattrup could be a candidate for thermonet or individual heat pumps in the future (*Varmeplan Horsens Kommune*).*

**Varmeplan 2023-2030 – for Horsens Kommune, last revised 08.06.23, not legally binding.*



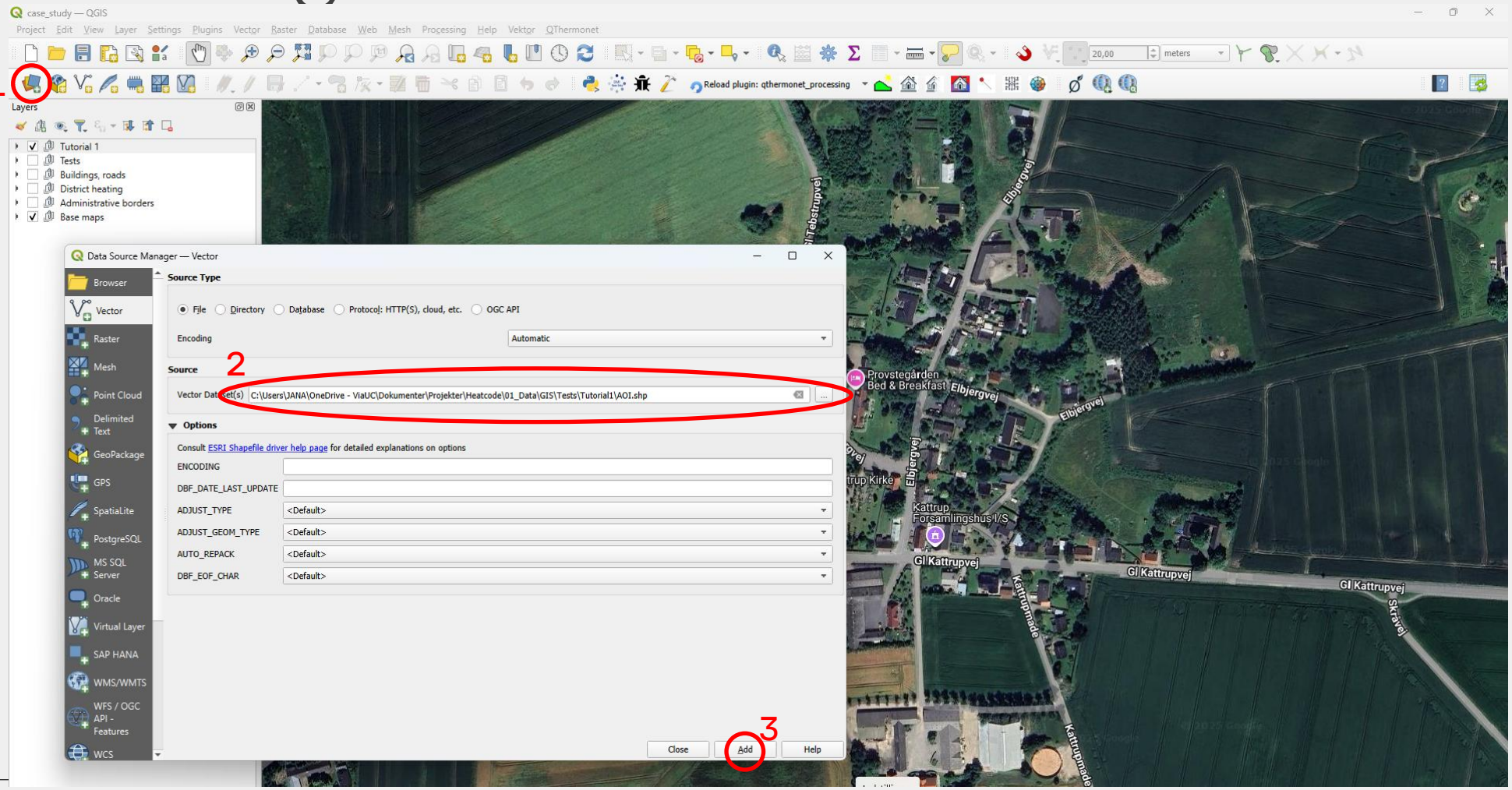


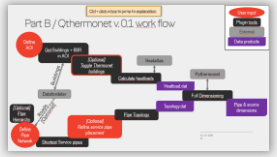
Click to return to overview



Part B1 / Buildings and heat loads 1

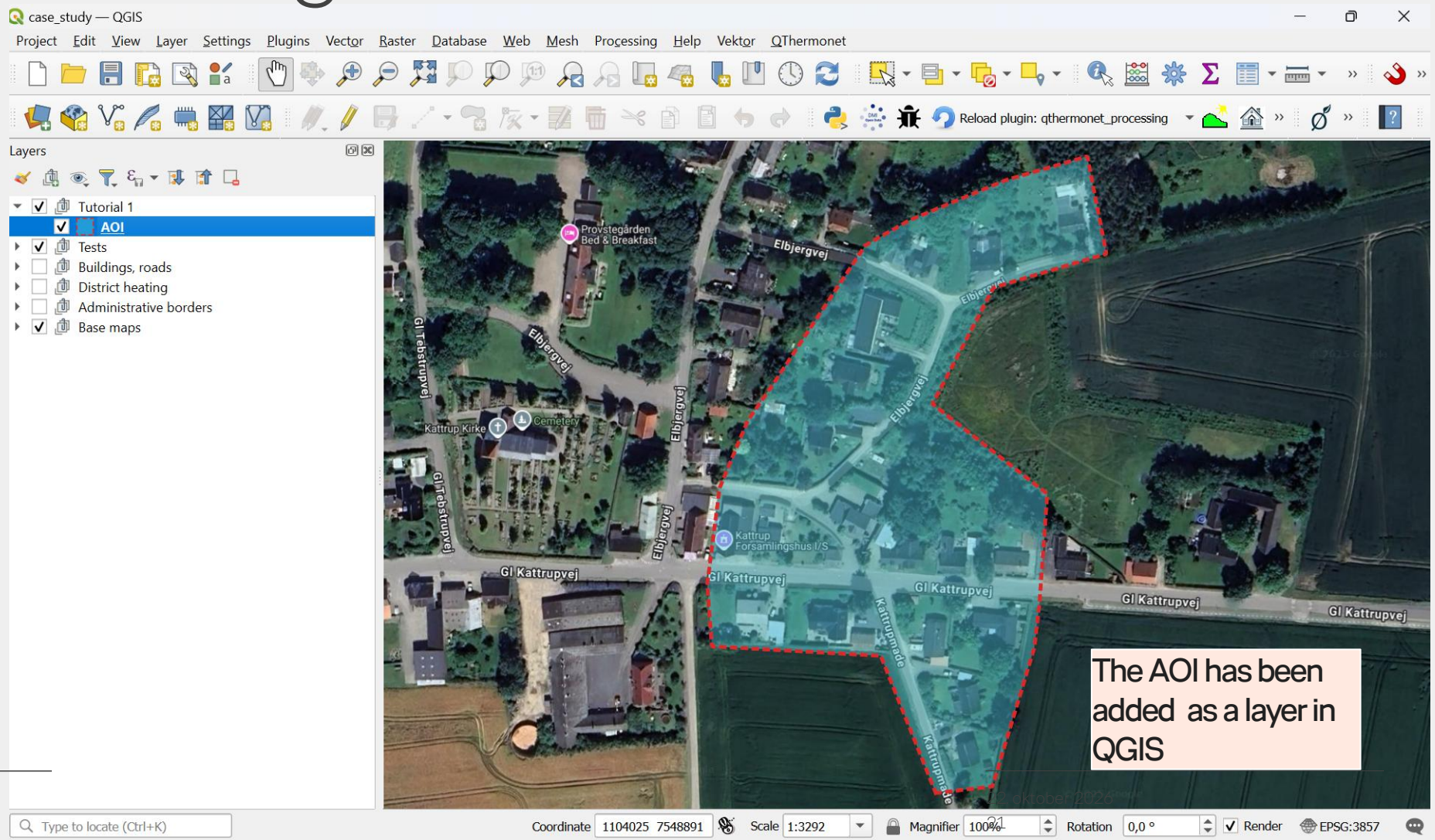
We define the area of interest (AOI) for the thermonet in the eastern part of Kattrup using a single polygon in a shapefile/geojson file (distributed with this tutorial).

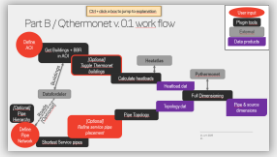




Part B1 / Buildings and heat loads 1

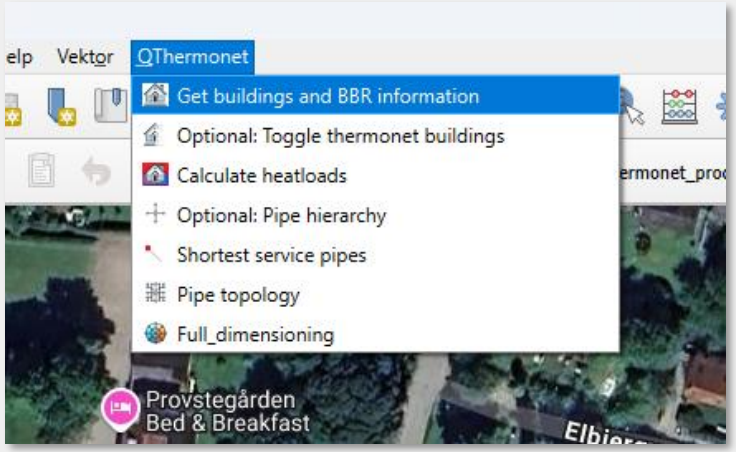
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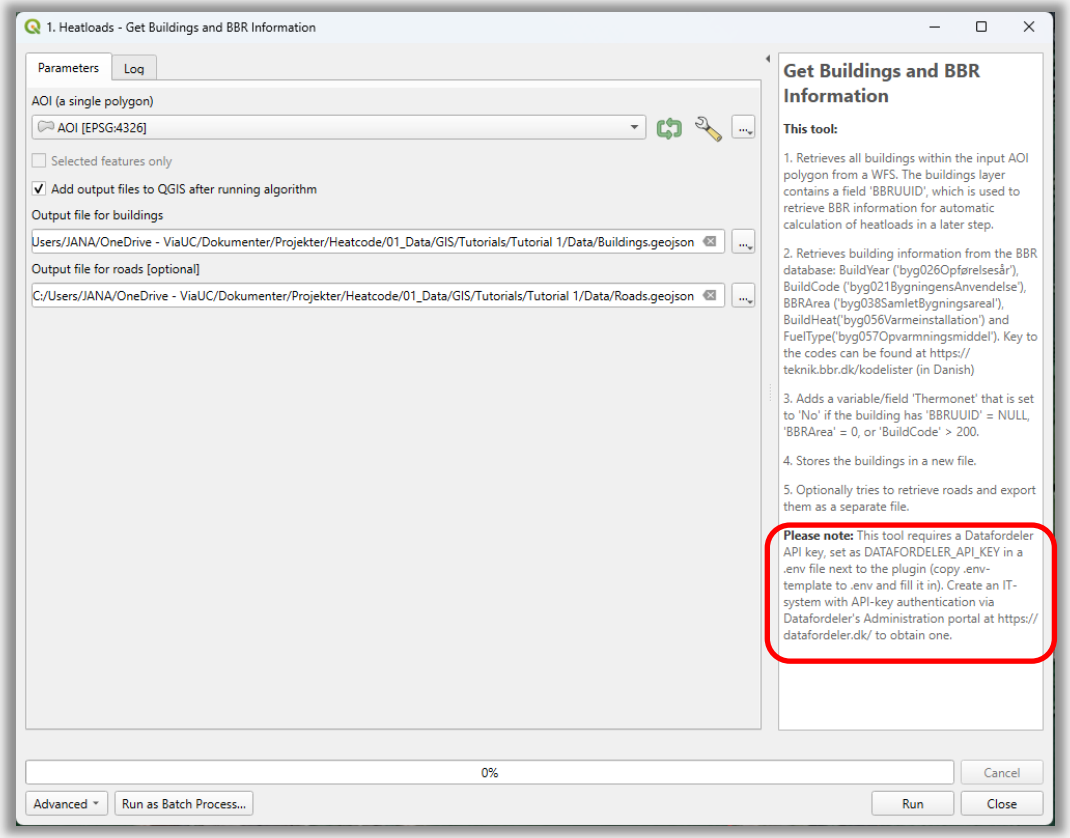




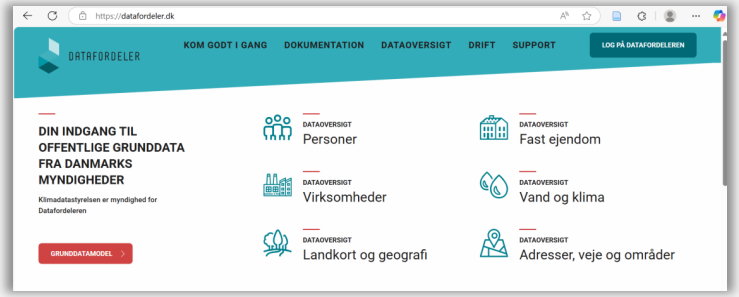
Part B1 / Buildings and heat loads 2

Using the first tool in the QThermonet plugin: ***Get Buildings and BBR information*** we retrieve the buildings (and optionally roads) within our AOI and the BBR information of the buildings (Total Area, Building Code, Construction Year).

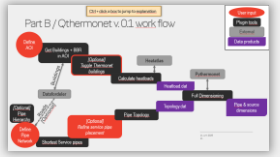




Running the tool requires an API key for Datafordeler.dk (Instructions to set up a user)



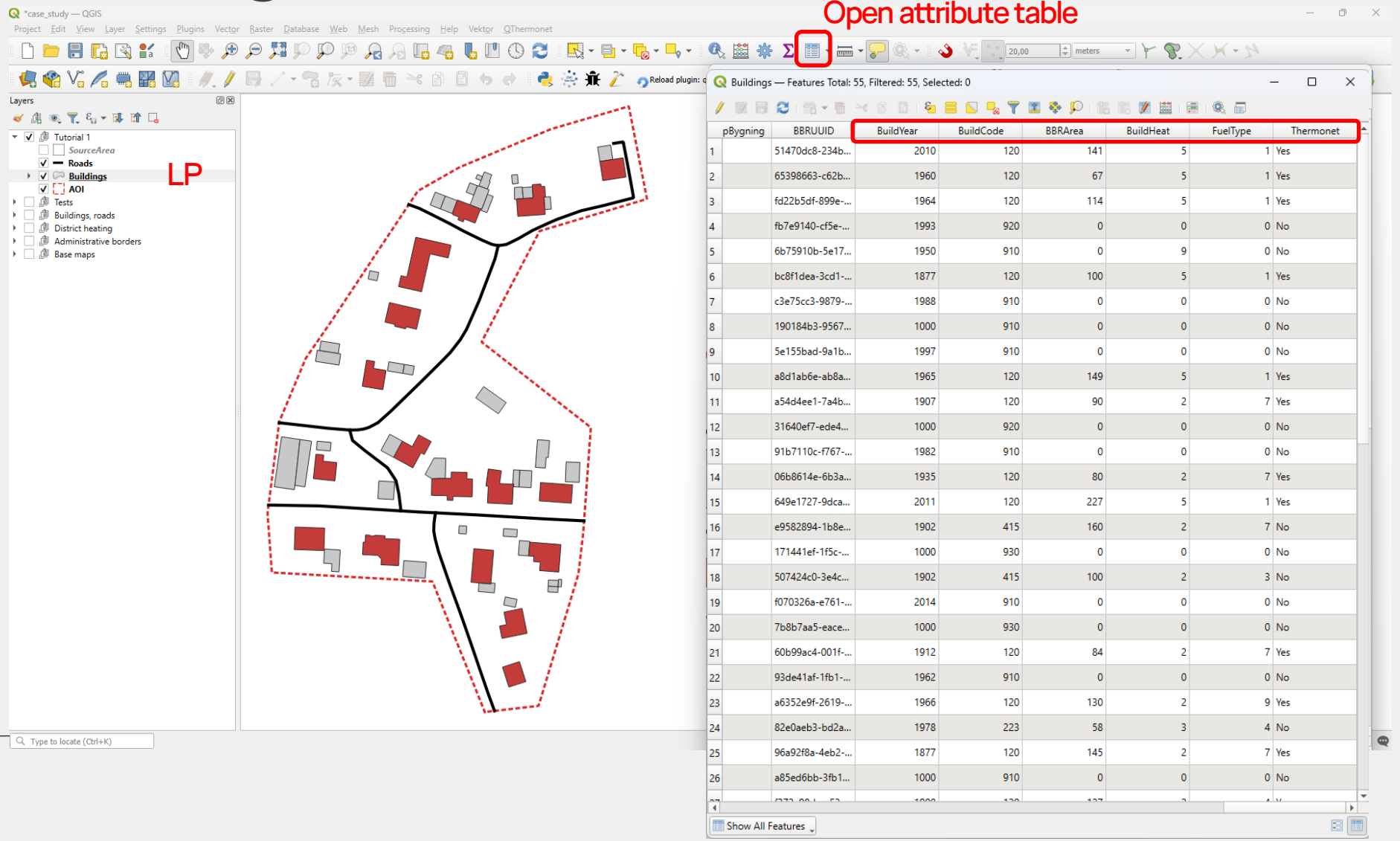
Part B1 / Buildings and heat loads 2



After running the tool, buildings (+ optionally roads) within the AOI are loaded into the layers panel (LP) of QGIS.

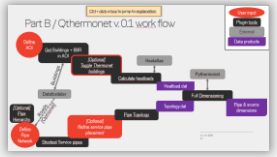
The **Buildings** table can be inspected using the **Open attribute table**.

The attribute table contains the data retrieved from Datafordeler + five columns from the BBR database + a column called **Thermonet**



	pBygning	BBRUUID	BuildYear	BuildCode	BBRArea	BuildHeat	FuelType	Thermonet
1		51470dc8-234b...	2010	120	141	5	1	Yes
2		65398663-c62b...	1960	120	67	5	1	Yes
3		fd22b5df-899e-...	1964	120	114	5	1	Yes
4		fb7e9140-cf5e-...	1993	920	0	0	0	No
5		6b75910b-5e17...	1950	910	0	9	0	No
6		bc8f1dea-3cd1-...	1877	120	100	5	1	Yes
7		c3e75cc3-9879-...	1988	910	0	0	0	No
8		190184b3-9567...	1000	910	0	0	0	No
9		5e155bad-9a1b...	1997	910	0	0	0	No
10		a8d1ab6e-ab8a...	1965	120	149	5	1	Yes
11		a54d4ee1-7a4b...	1907	120	90	2	7	Yes
12		31640ef7-ede4...	1000	920	0	0	0	No
13		91b7110c-f767-...	1982	910	0	0	0	No
14		06b8614e-6b3a...	1935	120	80	2	7	Yes
15		649e1727-9dca...	2011	120	227	5	1	Yes
16		e9582894-1b8e...	1902	415	160	2	7	No
17		171441ef-1f5c-...	1000	930	0	0	0	No
18		507424c0-3e4c...	1902	415	100	2	3	No
19		f070326a-e761-...	2014	910	0	0	0	No
20		7b8b7aa5-eace...	1000	930	0	0	0	No
21		60b99ac4-001f-...	1912	120	84	2	7	Yes
22		93de41af-1fb1-...	1962	910	0	0	0	No
23		a6352e9f-2619-...	1966	120	130	2	9	Yes
24		82e0aeb3-bd2a...	1978	223	58	3	4	No
25		96a92f8a-4eb2-...	1877	120	145	2	7	Yes
26		a85ed6bb-3fb1...	1000	910	0	0	0	No

Part B1 / Buildings and heat loads 2



BuildYear: Construction year of the building.

BuildCode: Type of building (residential/industry etc), see codelist here (in Danish): [Kodelister - BBR Teknik]

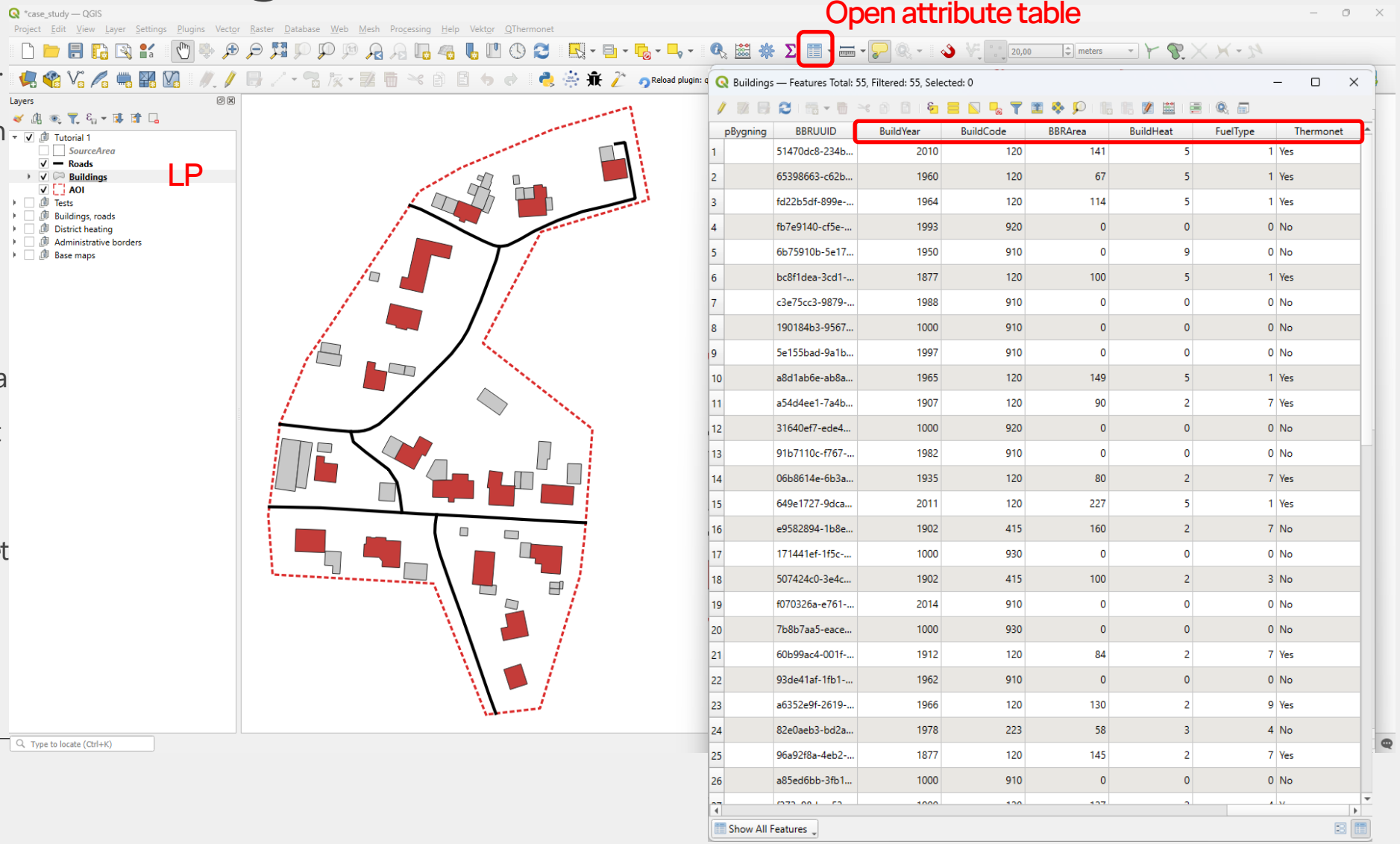
BBRarea: Total area of the building (including potential multiple floors)

BuildHeat: Heat supply type (BBR code)

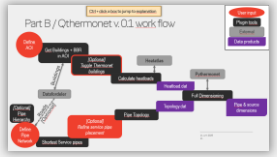
FuelType: Gas, oil, electricity (BBR code)

Thermonet: This column controls whether a building is included in the heat load calculation and connected to the thermonet or not. A building is by default included (set to 'Yes') if it has a BBRUUID, an area >0, and a BuildCode < 200 (residential).

Buildings in red on the map have Thermonet = 'Yes'



	pBygning	BBRUUID	BuildYear	BuildCode	BBRArea	BuildHeat	FuelType	Thermonet
1		51470dc8-234b...	2010	120	141	5	1	Yes
2		65398663-c62b...	1960	120	67	5	1	Yes
3		fd22b5df-899e-...	1964	120	114	5	1	Yes
4		fb7e9140-cf5e-...	1993	920	0	0	0	No
5		6b75910b-5e17...	1950	910	0	9	0	No
6		bc8f1dea-3cd1-...	1877	120	100	5	1	Yes
7		c3e75cc3-9879-...	1988	910	0	0	0	No
8		190184b3-9567...	1000	910	0	0	0	No
9		5e155bad-9a1b...	1997	910	0	0	0	No
10		a8d1ab6e-ab8a...	1965	120	149	5	1	Yes
11		a54d4ee1-7a4b...	1907	120	90	2	7	Yes
12		31640ef7-ede4...	1000	920	0	0	0	No
13		91b7110c-f767-...	1982	910	0	0	0	No
14		06b8614e-6b3a...	1935	120	80	2	7	Yes
15		649e1727-9dca...	2011	120	227	5	1	Yes
16		e9582894-1b8e...	1902	415	160	2	7	No
17		171441ef-1f5c-...	1000	930	0	0	0	No
18		507424c0-3e4c...	1902	415	100	2	3	No
19		f070326a-e761-...	2014	910	0	0	0	No
20		7b8b7aa5-eace...	1000	930	0	0	0	No
21		60b99ac4-001f-...	1912	120	84	2	7	Yes
22		93de41af-1fb1-...	1962	910	0	0	0	No
23		a6352e9f-2619-...	1966	120	130	2	9	Yes
24		82e0aeb3-bd2a...	1978	223	58	3	4	No
25		96a92f8a-4eb2-...	1877	120	145	2	7	Yes
26		a85ed6bb-3fb1...	1000	910	0	0	0	No

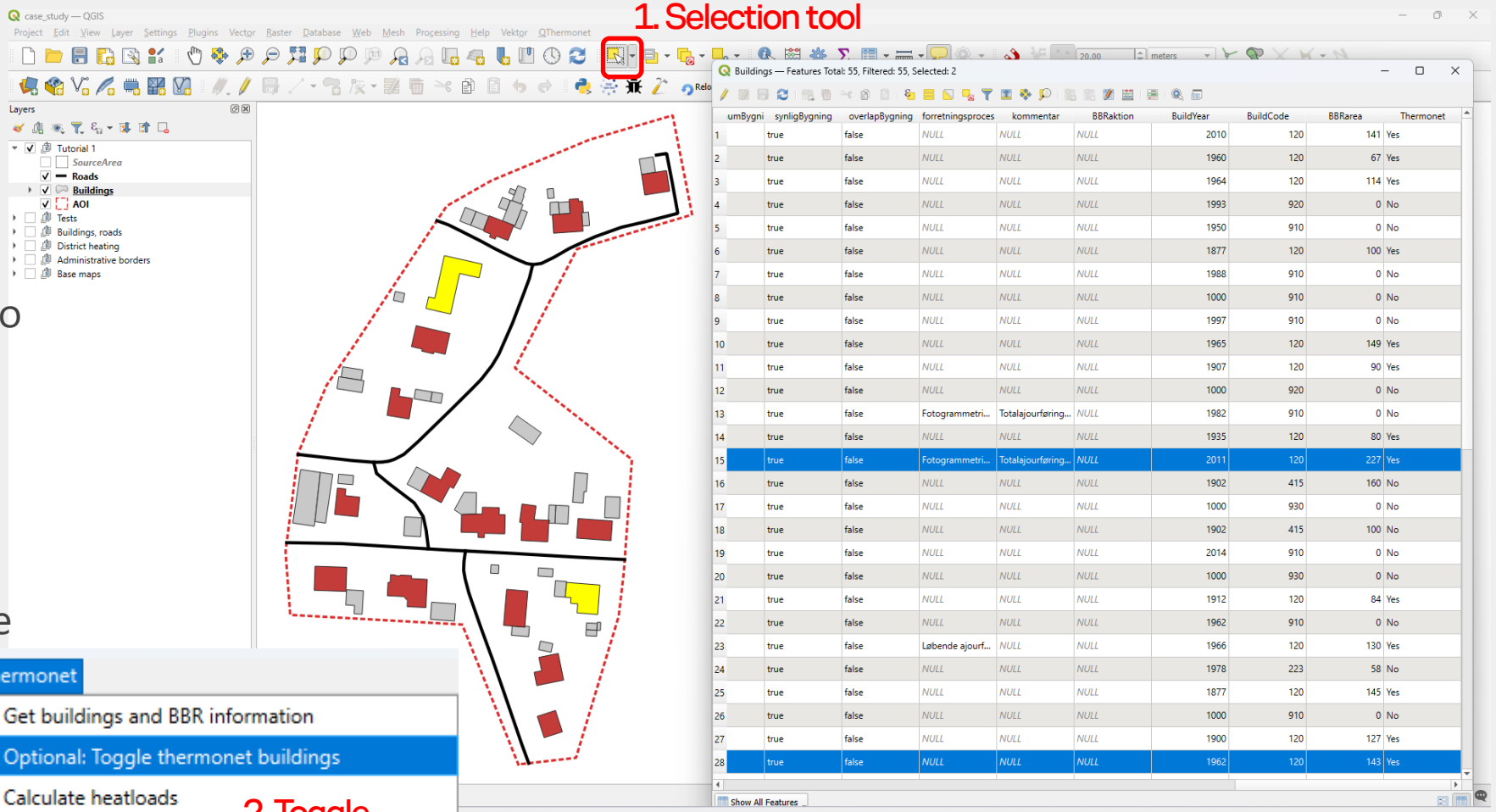


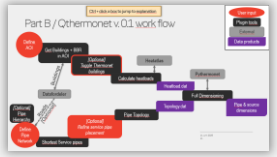
Part B1 / Buildings and heat loads 3

Optional: Toggle thermonet buildings

This is a helper tool that allows you to easily include/exclude buildings in the heat load and thermonet calculations.

1. Select the buildings that you want to toggle on/off the thermonet using the selection tool (they turn blue in the attribute table). Hold down 'ctrl' to select multiple buildings.
2. Open the **Optional: Toggle thermonet buildings** tool from the QThermonet drop-down menu.

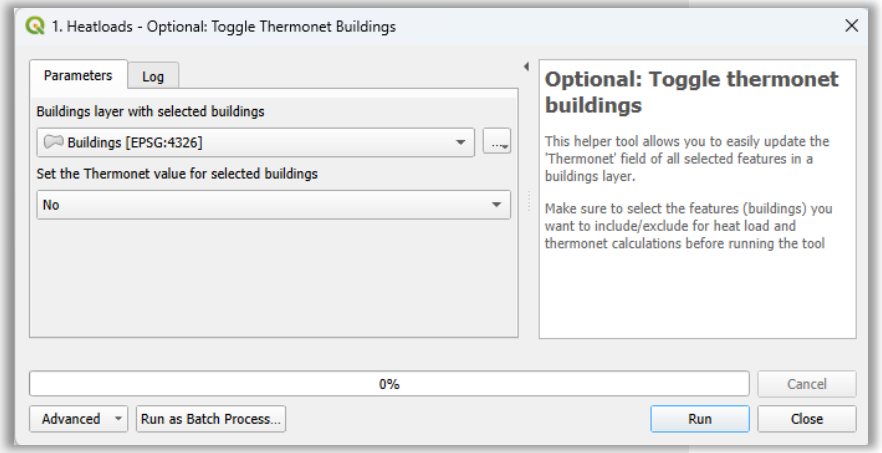


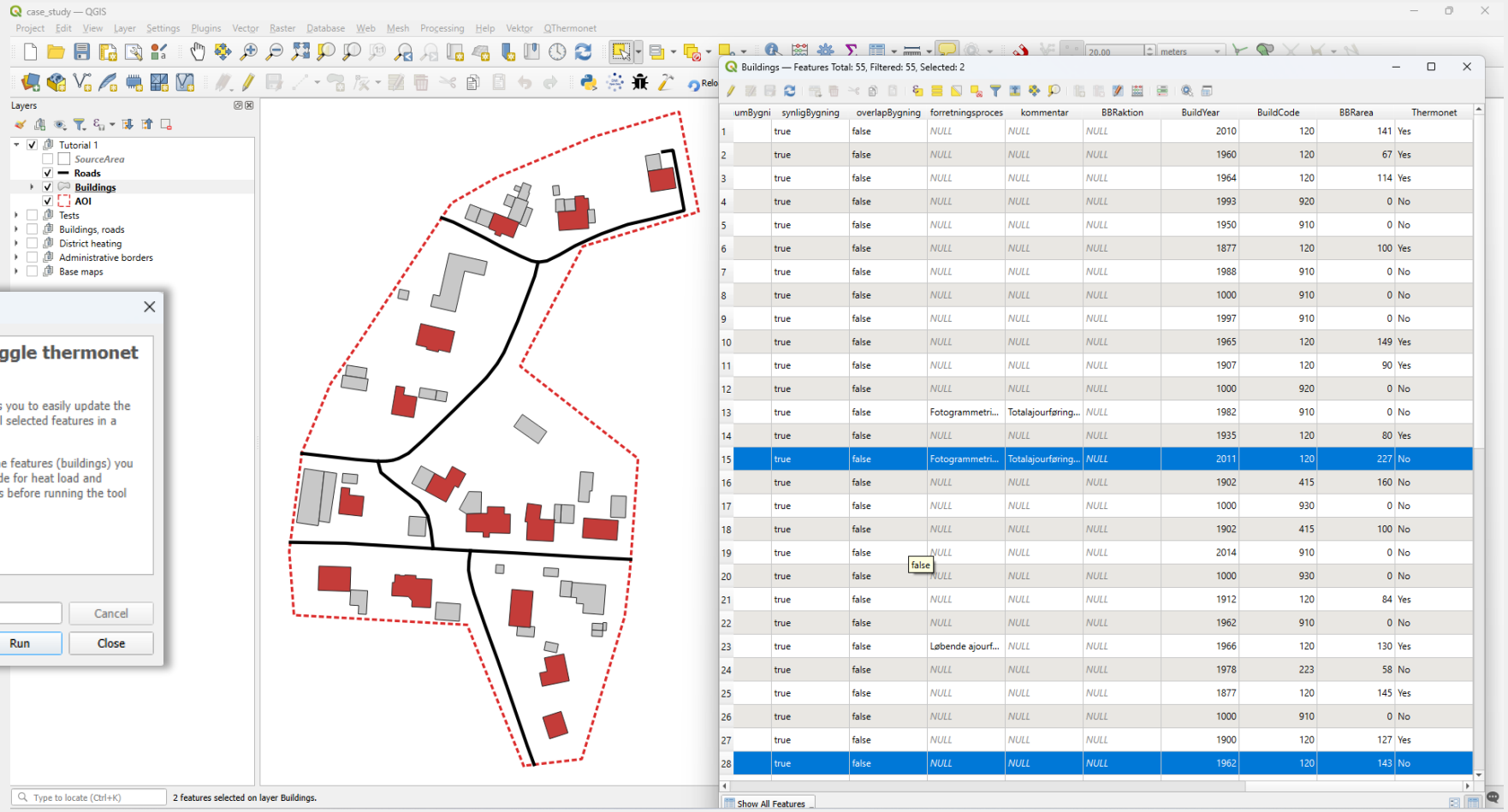


Part B1 / Buildings and heat loads 3

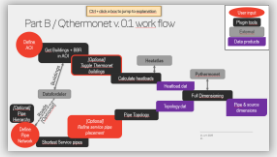
Optional: Toggle thermonet buildings

3. Run the tool to toggle off the selected buildings (sets 'Thermonet' to 'No').





Alternative: Update the attribute table directly.

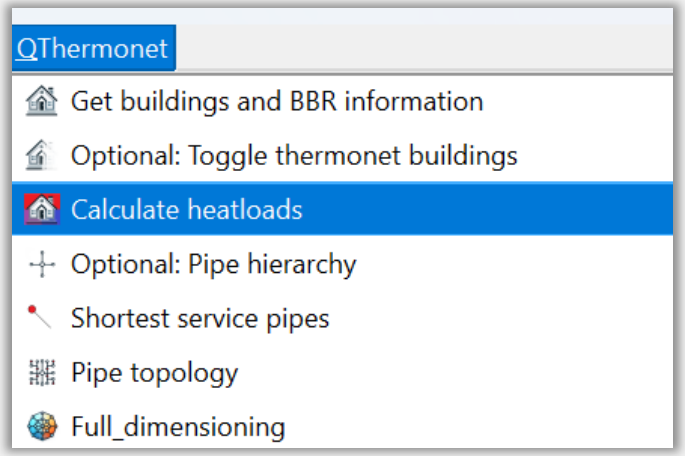


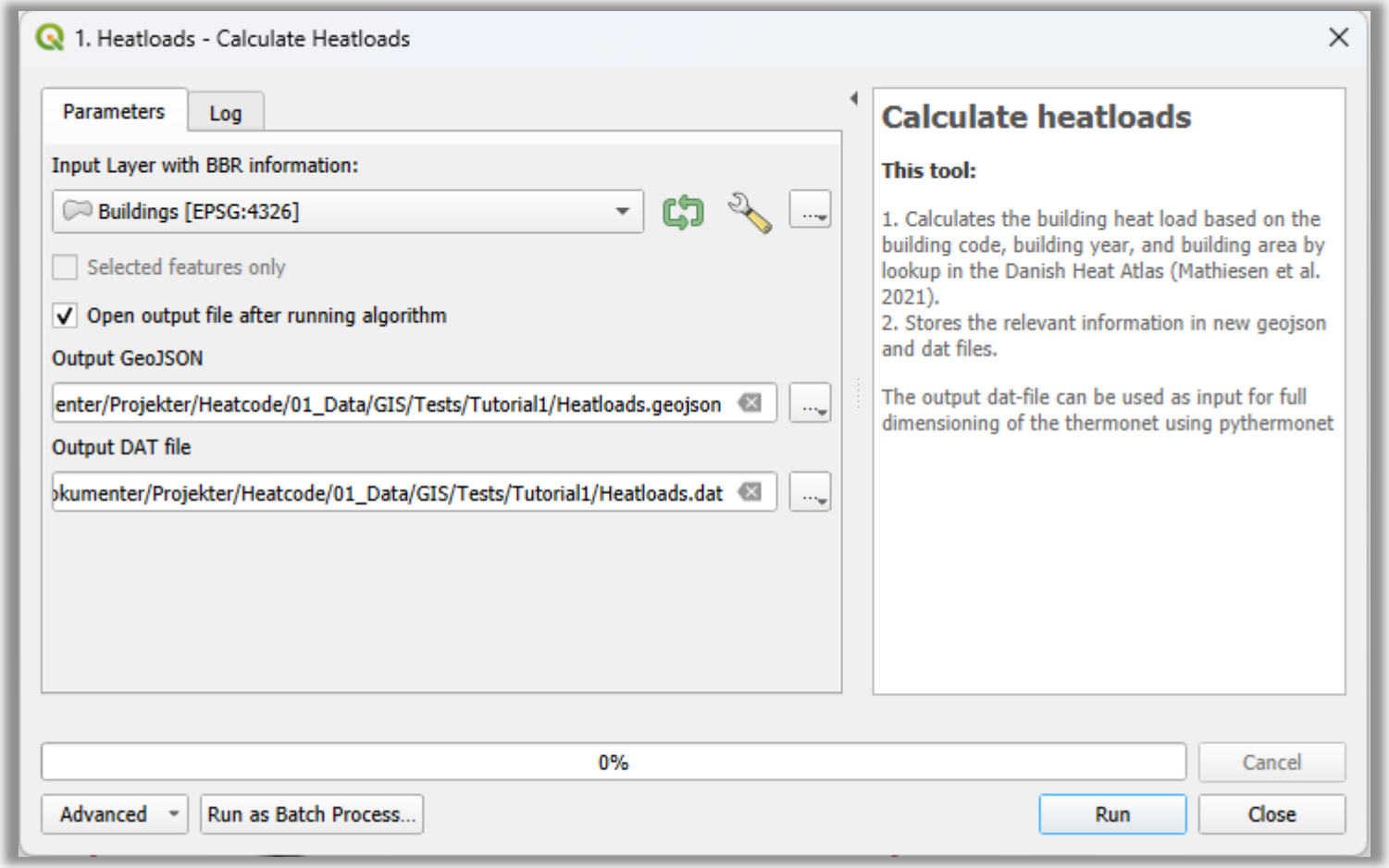
Part B1 / Buildings and heat loads 4

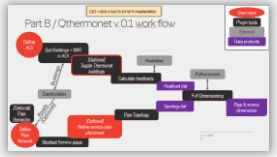
Calculate Heatloads

This tool calculates annual, winter, and maximum daily heat load for a building based on the BBR information.

Creates the heatload input file (.dat) required for running pythermonet dimensioning







Part B1 / Buildings and heat loads 4

Calculate Heatloads

- The tool looks up the annual heat load (kWh/m2) for each building in the thermonet based on the Danish Heat Atlas (2019)
- Multiplies by building area to get total load.
- For now*, multiplies by a fixed factor to get winter/peak daily heat load:

Multiplication factors:

Winter heat load = 1.6 x annual heat load (per time)

Daily heat load = 3.2 x winter heat load (per time)

**To be updated in future versions*

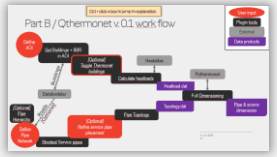
Specifikt årligt varmeforbrug i kWh/m² for forskellige bygningsanvendelseskoder samt alderskategorier.

Construction year

Building Codes

Anvendelse	<1850	1850-1930	1931-1950	1951-1960	1961-1972	1973-1978	1979-1998	1999-2006	2007<
110	137	156	173	179	138	126	115	106	82
120	152	185	197	163	123	110	97	82	65
121	152	185	197	163	123	110	97	82	65
130	170	180	192	172	130	112	80	69	67
131	170	180	192	172	130	112	80	69	67
132	170	180	192	172	130	112	80	69	67
140	143	139	144	148	117	116	84	76	68
150	182	177	164	141	128	180	122	111	86
160	249	206	171	186	153	143	125	112	82
185	142	172	196	155	151	131	106	74	83
190	142	172	196	155	151	131	106	74	83
210	215	244	235	190	198	192	157	166	148
211	0	0	0	0	0	0	0	0	0
212	0	0	0	0	0	0	0	0	0

Mathiesen, B. V., Lund, H., Nielsen, S., Sorknæs, P., Moreno, D., & Thellufsen, J. Z. (2021). *Varmeplan Danmark 2021 - Baggrundsrapport*. Aalborg Universitet.



Part B1 / Buildings and heat loads 4

Calculate Heatloads

Note: for now, the cooling loads are set to 0*

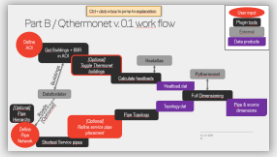
**To be updated in future versions*

Heatloads.dat

Fil Rediger Vis

Heat_pump_ID	Yearly_heating_load_(W)	Winter_heating_load_(W)	Daily_heating_load_(W)	Year_COP	Winter_COP	Hour_COP	dT_HP_Heating
	Yearly_cooling_load_(W)	Summer_cooling_load_(W)	Daily_cooling_load_(W)	EER	dT_HP_Cooling		
1	1046	1673	5354	3.3	3.4	3.0	3.0	0	0	0	0.0	0.0
2	1246	1993	6378	3.3	3.4	3.0	3.0	0	0	0	0.0	0.0
3	1600	2559	8189	3.3	3.4	3.0	3.0	0	0	0	0.0	0.0
6	2110	3377	10806	3.3	3.4	3.0	3.0	0	0	0	0.0	0.0
10	2091	3345	10704	3.3	3.4	3.0	3.0	0	0	0	0.0	0.0
11	1899	3039	9725	3.3	3.4	3.0	3.0	0	0	0	0.0	0.0
14	1798	2877	9206	3.3	3.4	3.0	3.0	0	0	0	0.0	0.0
21	1773	2836	9075	3.3	3.4	3.0	3.0	0	0	0	0.0	0.0
23	1824	2919	9341	3.3	3.4	3.0	3.0	0	0	0	0.0	0.0
25	3060	4896	15667	3.3	3.4	3.0	3.0	0	0	0	0.0	0.0
27	2680	4288	13722	3.3	3.4	3.0	3.0	0	0	0	0.0	0.0
31	2258	3613	11562	3.3	3.4	3.0	3.0	0	0	0	0.0	0.0
35	1604	2566	8211	3.3	3.4	3.0	3.0	0	0	0	0.0	0.0
51	2279	3647	11670	3.3	3.4	3.0	3.0	0	0	0	0.0	0.0
52	1333	2133	6826	3.3	3.4	3.0	3.0	0	0	0	0.0	0.0

Ln 17, Col 1 | 938 tegn | Almindelig tekst | 100% | Windows (CRLF) | UTF-8



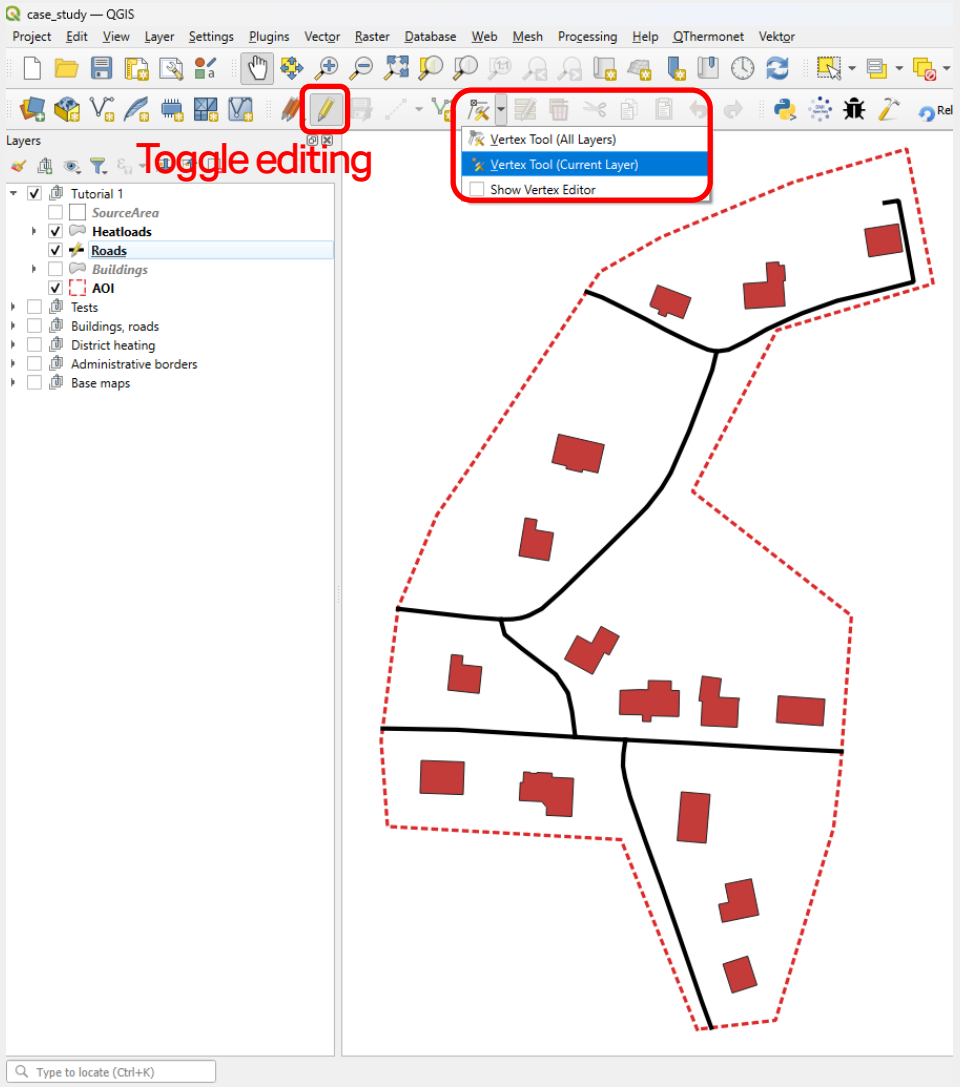
Part B2 / Thermonet pipe layout 1

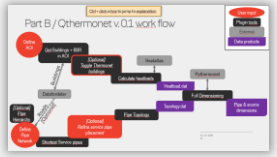
Define pipe network

In this step we will modify the roads network retrieved from the tool **Get buildings and BBR information** to define the main pipe network for our thermonet.

Please note that the pipe network needs to have a **tree structure** (i.e. no loops) in order to work.

To modify the network click the **Toggle editing** button in QGIS and use the **Vertex Tool (current layer)**.





Part B2 / Thermonet pipe layout 1

Define pipe network

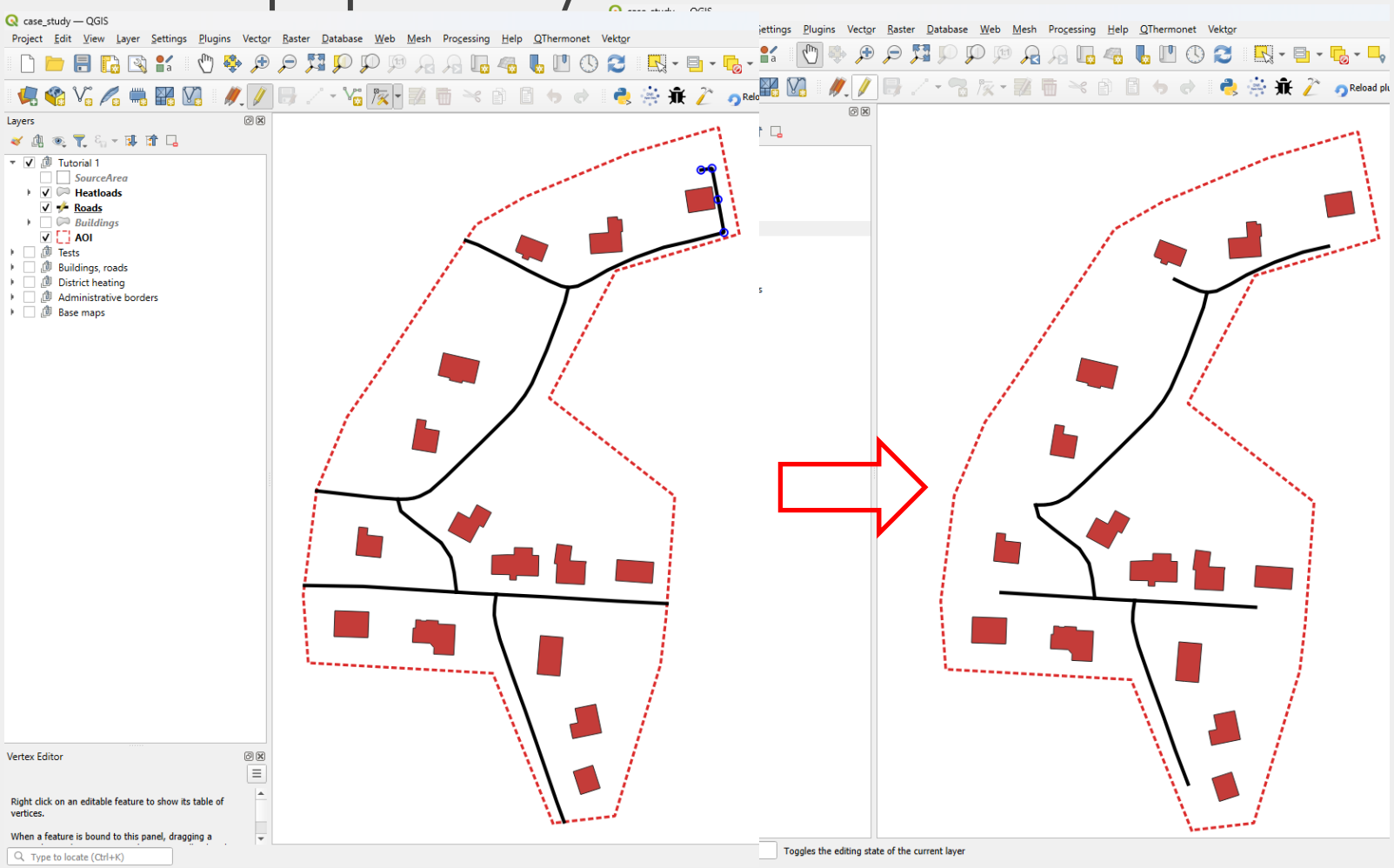
With the vertex tool selected, you can click, drag or delete vertices of the road network.

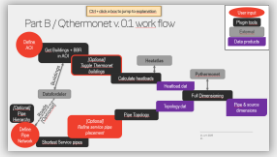
Remove unnecessary road segments to create your main pipe network.

Don't connect the houses to the network yet!

Save your edits and toggle off the layer by clicking the toggle editing button again.

Note: Your pipe network does not need to follow existing road network, you can draw your own network e.g. in places where it would be cheaper to dig.

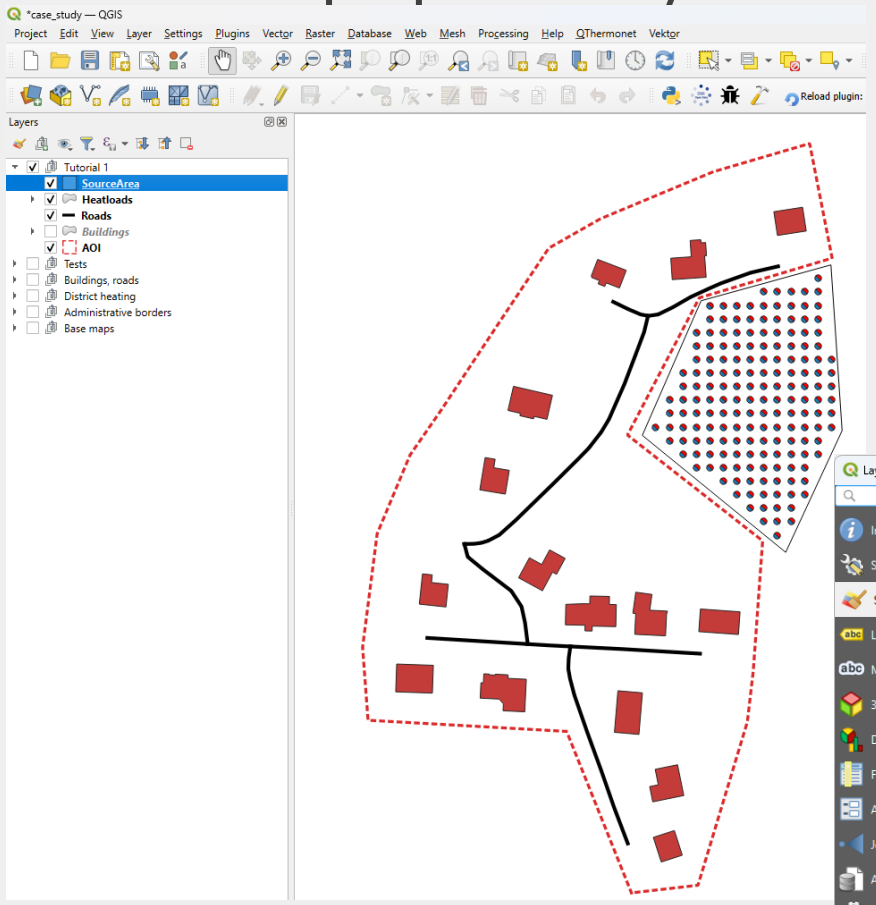


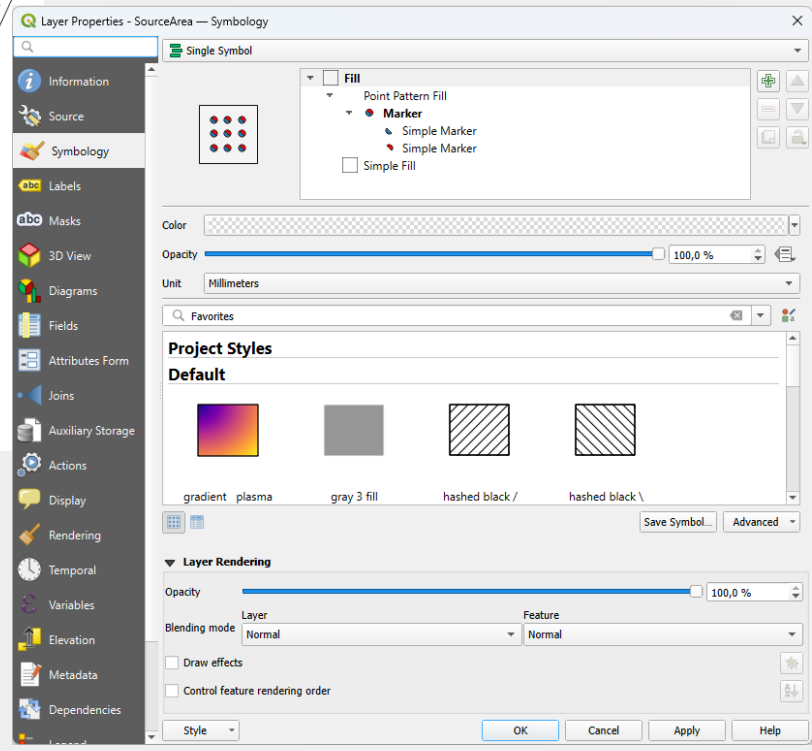


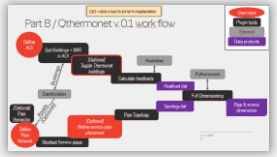
Part B2 / Thermonet pipe layout 2

Pipe Hierarchy

- To define the pipes hierarchy (main/peripheral pipes) you need to decide on the location of your source area (the location of borefield or horizontal heat exchangers).
- Load the **SourceArea.shp** file into QGIS using the data source manager like you did for the AOI (link). The source area is defined as a single polygon.
- *Optional:* set the style of the layer by double-clicking the layer and changing the 'symbology'.





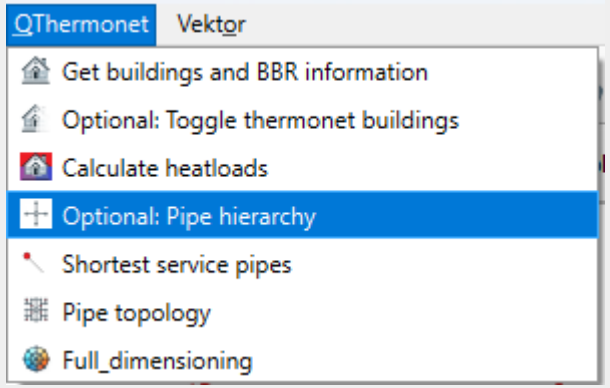


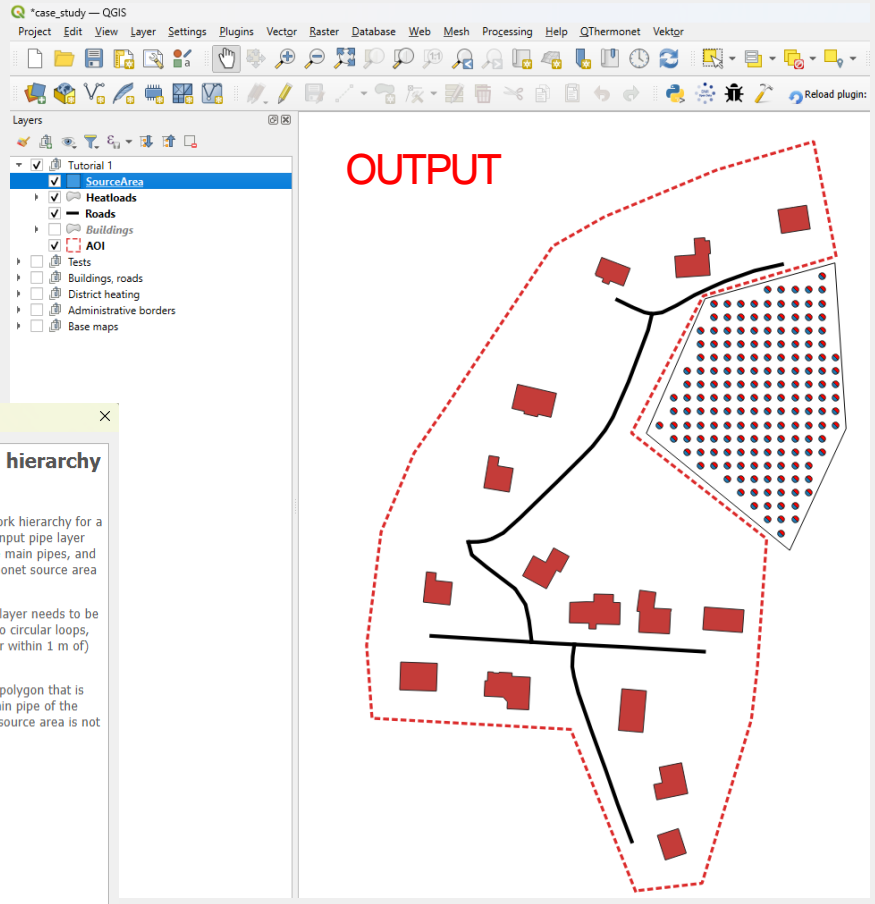
Part B2 / Thermonet pipe layout 2

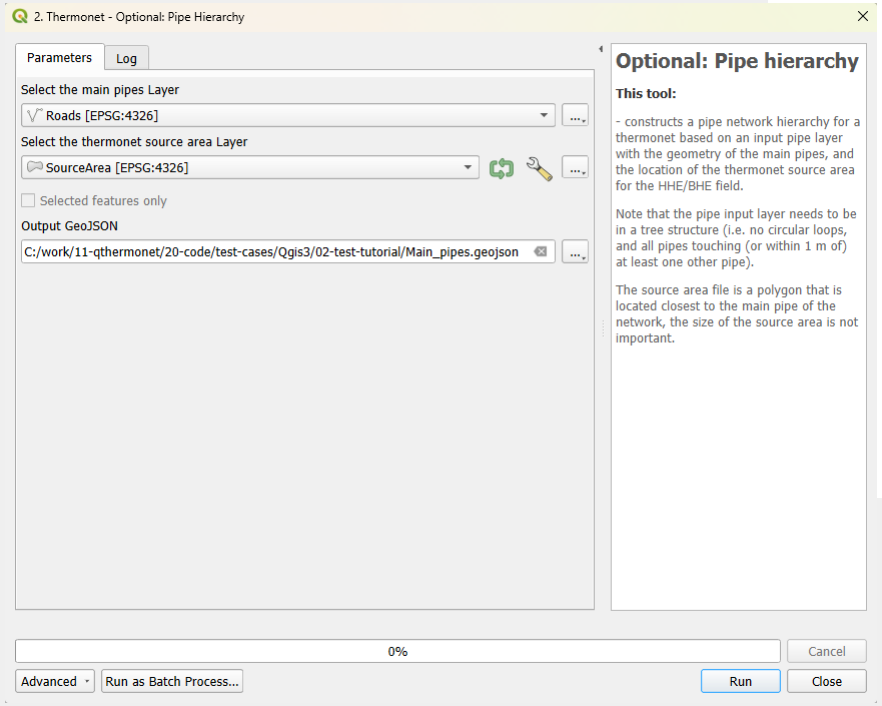
Pipe Hierarchy

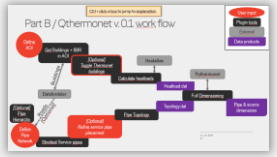
- Next, run the **Optional: Pipe hierarchy** tool from the QThermonet plugin menu.
- The tool creates a new output layer (**Pipes**) that contains a field **Level**
- The highest level ("0") is assigned to the main pipe closest to the sourceArea, branching pipes are assigned gradually lower levels ("+1")

- *Note: Alternatively the pipe level field can be added and populated manually in the attribute table of the roads file*







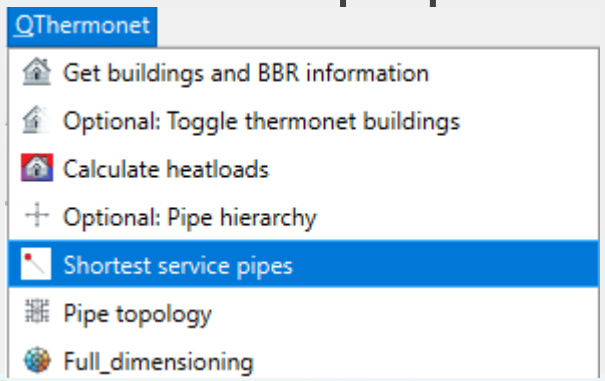


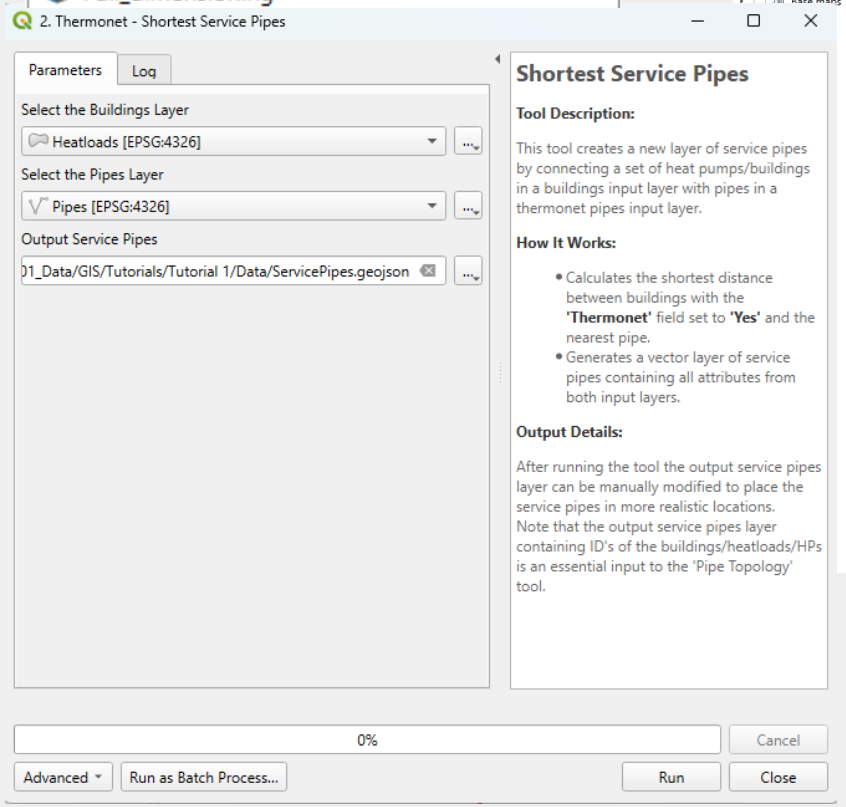
Part B2 / Thermonet pipe layout 3

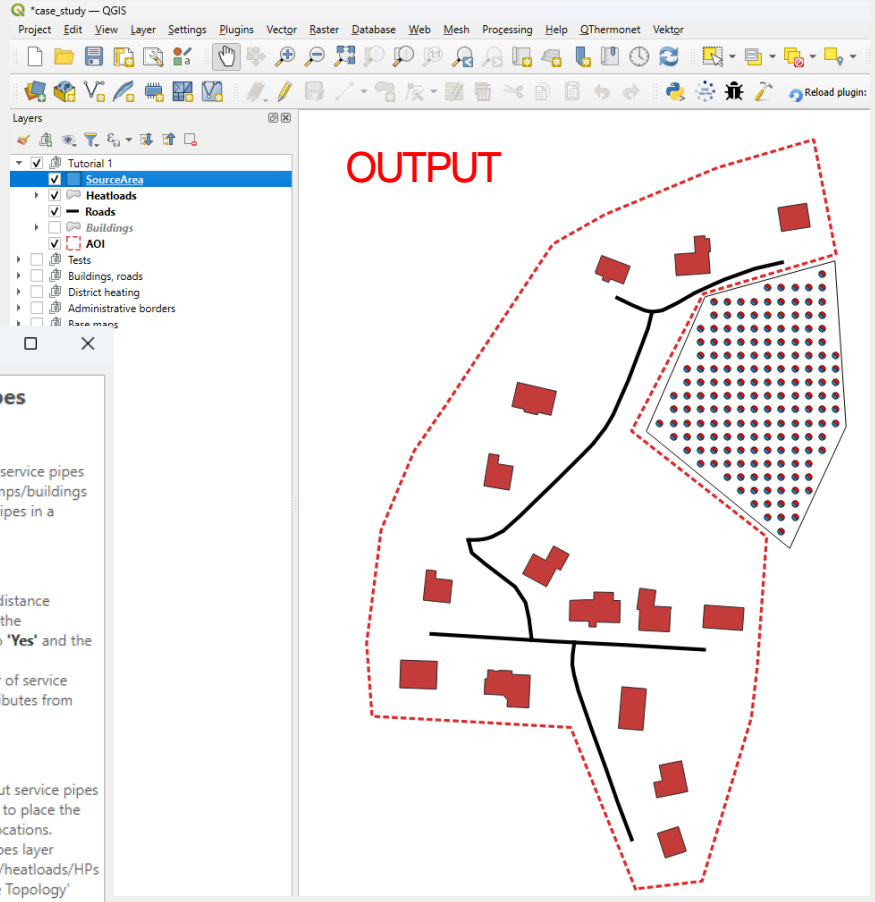
Shortest service pipes

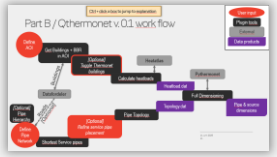
Now the main pipe network is complete, and it is time to combine the houses/heat pumps to the network with service pipes.

- The tool **Shortest service pipes** will create a new layer with service pipes connecting each house with the closest pipe segment.
- The output **ServicePipes** layer contains one line segment for each service pipe illustrated with a red dot at the location of the heat pump and a black line connecting to the main pipe network.
- Optional: The exact location of the service pipe can be refined by setting the ServicePipes layer in **Toggle editing** mode and using the **Vertex tool (current layer)** as was done for the main pipes layer (link)









Part B2 / Thermonet pipe layout 3

Shortest service pipes

Note that the attribute table of the ServicePipes layer contains the information from both the heat pumps and houses (BBR, heat loads, heat pump ID (id_lokalId) and the main pipe network (Level)).

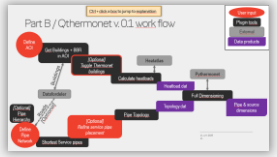
It therefore ties together the information from step B1 and B2 in this tutorial and provide the link between the thermonet and the heat loads.

NB.: If you run the tool again with the same file name, first remove the old ServicePipes layer from the project, or choose a new file name.

Service pipes — Features Total: 17, Filtered: 17, Selected: 0

	id_lokalId	BBRUUID	Thermonet	BuildYear	BuildCode	BBRArea	YrHeatLoad	WiHeatLoad	DyHeatLoad	ellip_length	Level
1	1038035605	51470dc8-234...	Yes	2010	120	141	1046	1673	5354	211,7	0
2	1038035696	65398663-c62...	Yes	1960	120	67	1246	1993	6378	111,8	2
3	1038035704	fd22b5df-899...	Yes	1964	120	114	1600	2559	8189	140,5	1
4	1038035713	bc8f1dea-3cd...	Yes	1877	120	100	2110	3377	10806	68,7	3
5	1038035720	a8d1ab6e-ab8...	Yes	1965	120	149	2091	3345	10704	140,5	1
6	1038035794	a54d4ee1-7a4...	Yes	1907	120	90	1899	3039	9725	140,5	1
7	1038037287	06b8614e-6b...	Yes	1935	120	80	1798	2877	9206	211,7	0
8	1069783393	649e1727-9dc...	Yes	2011	120	227	1683	2693	8618	45,5	1
9	1081328666	60b99ac4-001...	Yes	1912	120	84	1773	2836	9075	211,7	0
10	1081356087	a6352e9f-261...	Yes	1966	120	130	1824	2919	9341	111,8	2
11	1081363287	96a92f8a-4eb...	Yes	1877	120	145	3060	4896	15667	68,7	3
12	1081376105	f373a98d-ec5...	Yes	1900	120	127	2680	4288	13722	140,5	1
13	1081379538	2fd46534-086...	Yes	1962	120	143	2007	3210	10272	68,7	3
14	1081382535	3b842d62-c41...	Yes	1877	120	107	2258	3613	11562	68,7	3
15	1090189622	8e6e9de3-0da...	Yes	1880	120	76	1604	2566	8211	45,5	1
16	1102801346	2fdd2ca3-762...	Yes	1877	120	108	2279	3647	11670	211,7	0

Show All Features

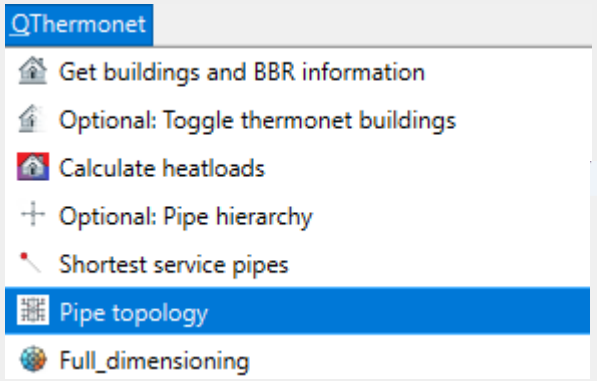


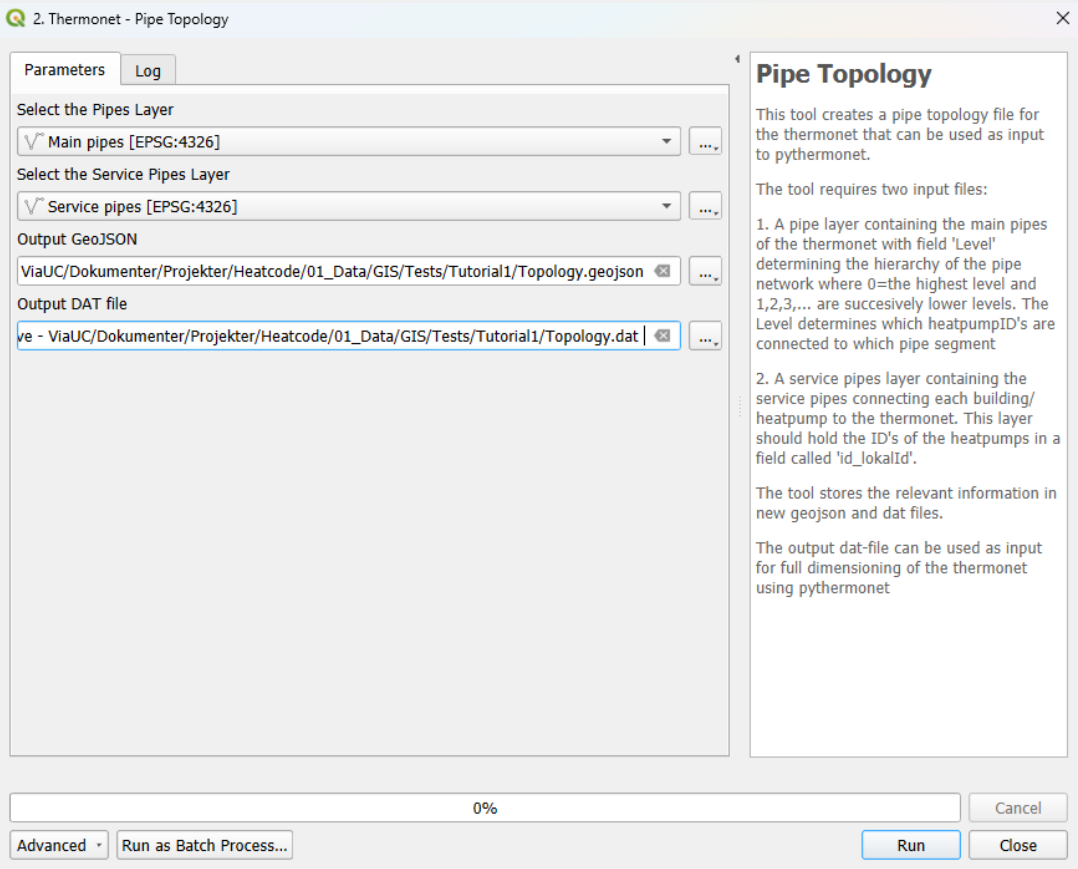
Part B2 / Thermonet pipe layout 4

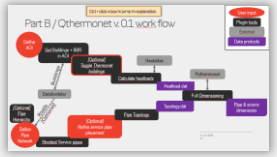
Pipe Topology

The pipe topology tool combines the pipe network and the service pipe network (that contains the handles/IDs of the heat pumps).

The tool generates the Topology.dat file which is required as input for the thermonet dimensioning.







Part B2 / Thermonet pipe layout 4

Pipe Topology

Output of the pipe topology tool.

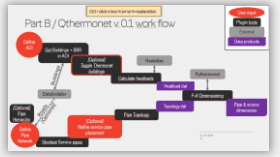
*Note: for now, the SDR is set to 17 and the maximum pressure loss to 180*Trace_length by default.*

Topology.dat

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```
Section SDR     Trace_(m)       Number_of_traces        Max_pressure_loss_(Pa)  HP_ID_vector
Service_pipe_1  17      18.79   1       3383    1038035605
Service_pipe_2  17      9.79    1       1761    1038035696
Service_pipe_3  17      10.04   1       1807    1038035704
Service_pipe_4  17      8.15    1       1466    1038035713
Service_pipe_5  17      11.88   1       2139    1038035720
Service_pipe_6  17      11.93   1       2147    1038035794
Service_pipe_7  17      10.47   1       1884    1038037287
Service_pipe_8  17      11.49   1       2068    1081328666
Service_pipe_9  17      10.18   1       1833    1081356087
Service_pipe_10 17      6.06    1       1091    1081363287
Service_pipe_11 17      5.56    1       1002    1081376105
Service_pipe_12 17      5.76    1       1036    1081382535
Service_pipe_13 17      5.80    1       1043    1090189622
Service_pipe_14 17      4.33    1       780     1102801346
Service_pipe_15 17      15.33   1       2759    1210916157
Pipe_branch_1   17      48.40   1       8712    1038035713, 1081363287, 1081382535
Pipe_branch_2   17      93.00   1       16740   1038035696, 1038035713, 1081356087, 1081363287, 1081382535, 1210916157
Pipe_branch_3   17      14.07   1       2532    1090189622
Pipe_branch_4   17      250.81  1       45146   1038035605, 1038035696, 1038035704, 1038035713, 1038035720, 1038035794,
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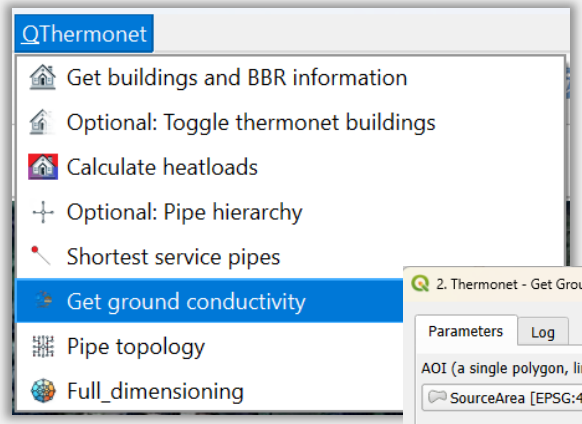
Ln 1, Col 1 | 1.122 tegn | 100% | Windows (CRLF) | UTF-8

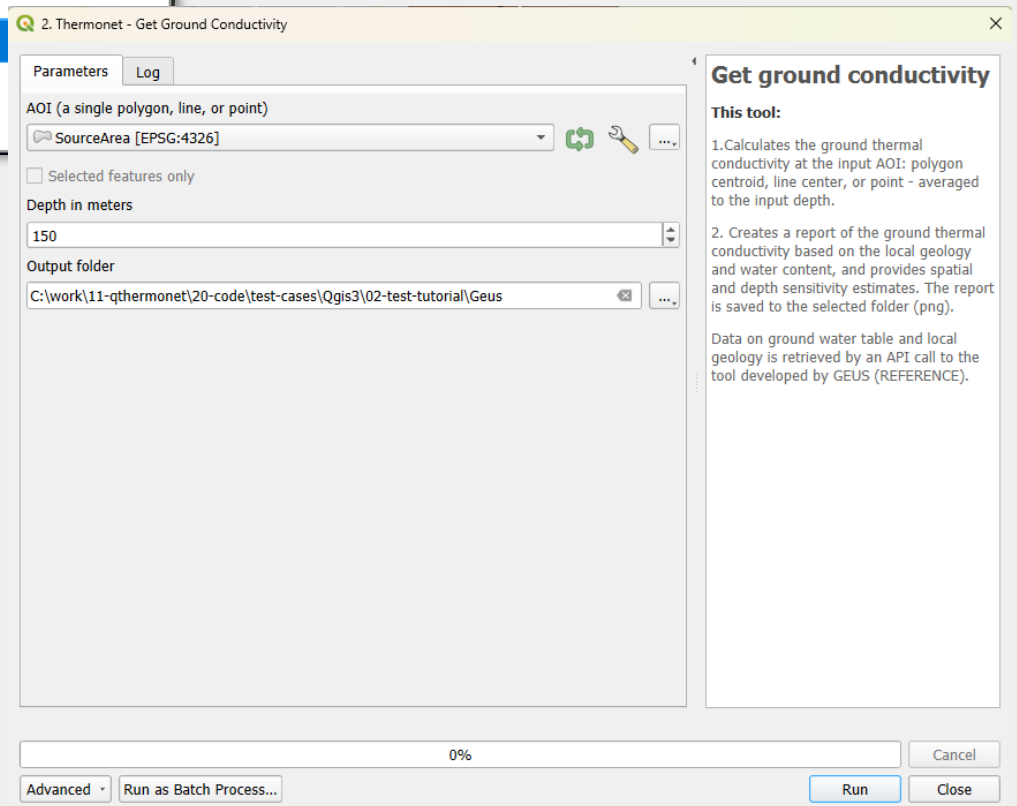


Part B2 / Ground thermal conductivity

(Optional)

To estimate the local ground thermal conductivity (GTC) in Denmark based on local geological and groundwater information you can use the **Get Ground Conductivity** tool

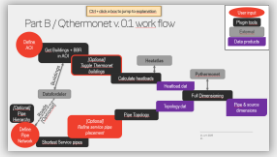




- Add the area-of-interest file containing a single polygon, polyline, or point centered on the area where you would like to estimate the GTC.
- Choose the depth in meters
- Select an output folder to save the GTC report (optional)

➢ The tool calls GEUS REST-service* to retrieve information on geology and groundwater level and calculate the depth-averaged GTC down to the specified depth

*Jordvarmedata og korttjenester | GEUS



Part B2 / Ground thermal conductivity

(Optional)

The tool relies on a database of measurements of thermal conductivities on Danish soils (Ditlevsen et al. 2014) [1].

The local geology is derived from FOHM which is a national hydrostratigraphic/geological model for Denmark [2].

The fraction of the depth profile that is above the water table is assumed to be unsaturated and is assigned a thermal conductivity of 1.0 W mK^{-1}.

The water table depth is derived from HIP-model [3].

Table 1. Thermal conductivity of some common, shallow, Danish sediments

Sediment type	Number of samples	Average thermal conductivity W mK^{-1}	Range W mK^{-1}	One standard deviation W mK^{-1}
Gyttja	3	0.68	0.58–0.86	0.15
Smectite-rich clay	3	0.98	0.80–1.14	0.17
Silty clay	10	1.15	0.90–1.42	0.17
Chalk*	4	1.62	1.49–1.80	0.13
Mica-rich, fine-grained sand	8	1.81	1.48–2.19	0.27
Till	19	1.89	1.40–2.66	0.30
Glacial sand, gravelly	4	2.24	1.98–2.43	0.19
Pure quartz sand	3	2.75	2.41–3.34	0.51

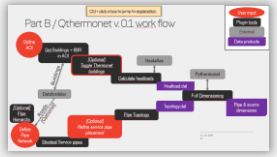
* Selected data from Balling *et al.* (1981).
Measurements were conducted with a needle probe (Hukseflux 2008) using water-saturated samples.

1. Ditlevsen et al., 2014. Estimating thermal conductivity from lithological descriptions – a new web-based tool for planning of ground-source heating and cooling. Geological Survey of Denmark and Greenland Bulletin 31, 55–58.

2. FOHM: GEUS – Grundvandsrapporter

3. HIP/DK-model:

*Jordvarmedata og korttjenester | GEUS



Part B2 / Ground thermal conductivity

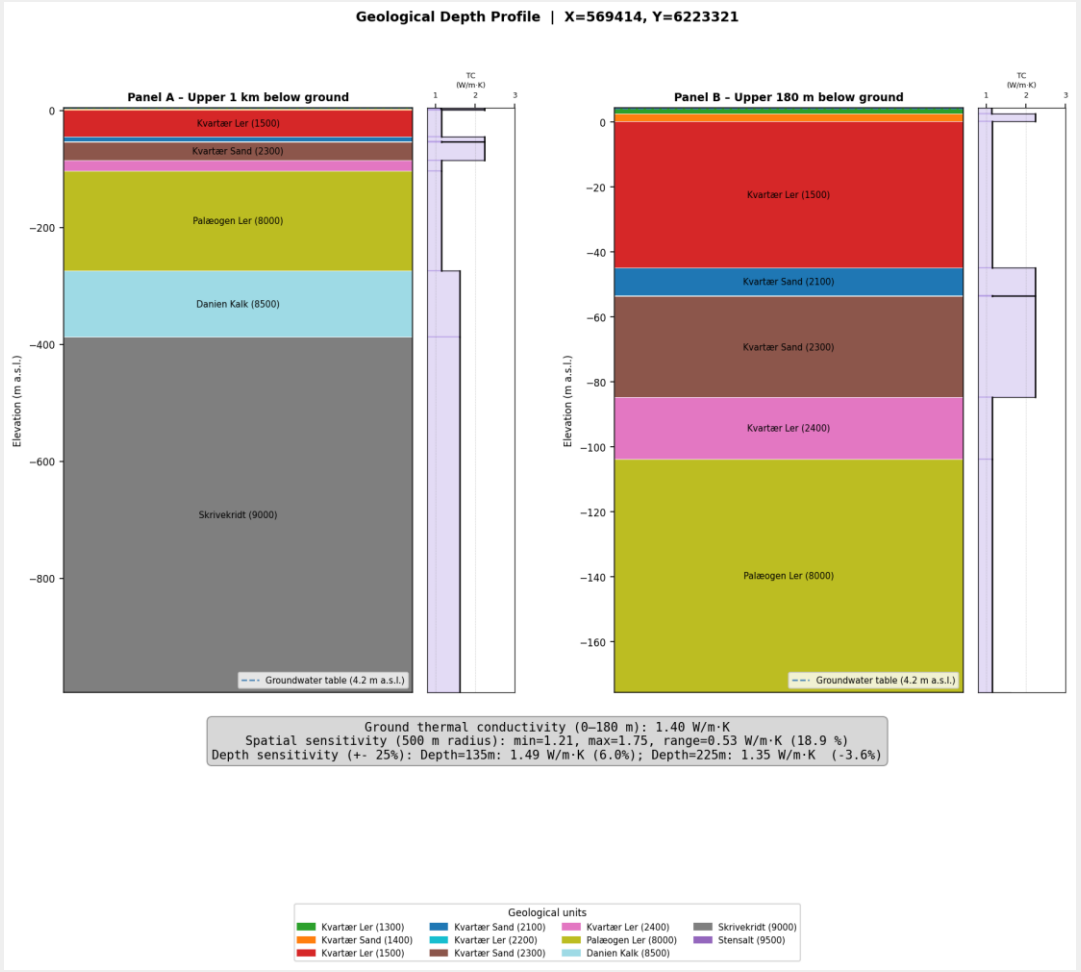
(Optional)

The output report shows the local geology at the polygon centroid, line midpoint, or point

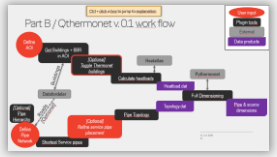
- The left panel shows the full geological profile with thick layers named (in Danish) and the water-corrected GTC depth profile
- The right panel shows the geological profile down to the selected depth and the corresponding water-corrected GTC depth profile
- The summary box provides the GTC and the spatial and depth sensitivities*
- A legend of all geological units is shown at the base

*The GTC spatial sensitivity is calculated in 8 points at a radius of 500 m from the centroid/point of interest. The min, max, and range is reported.

Depth sensitivity: GTC provided at +-25% depth.



*Jordvarmedata og korttjenester | GEUS

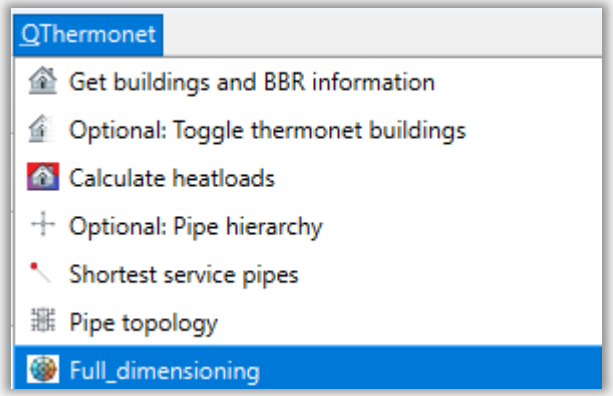


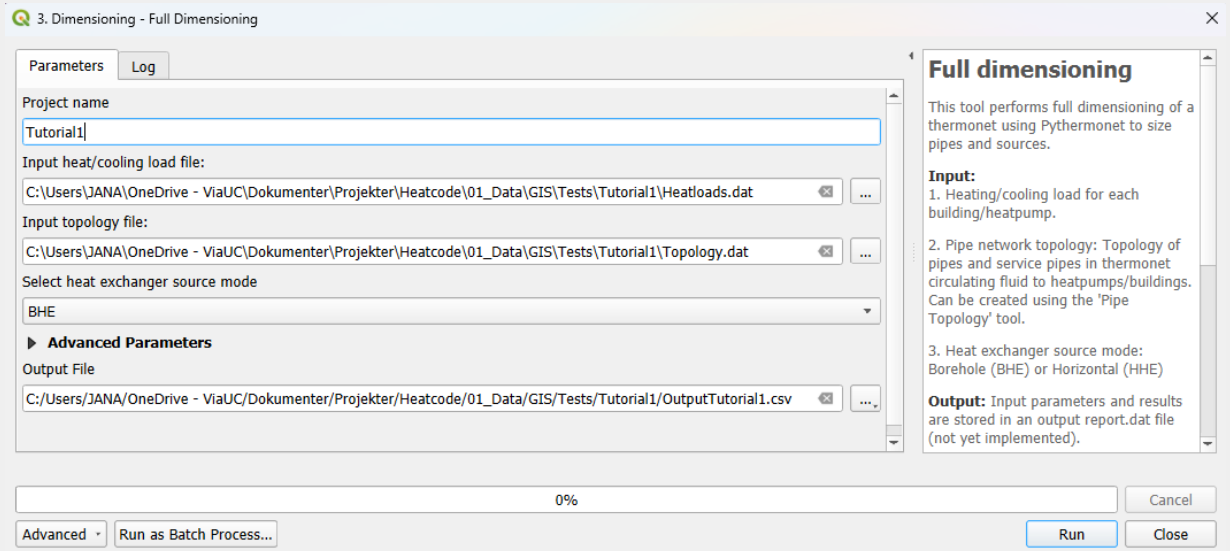
Part B3 / Full dimensioning 1

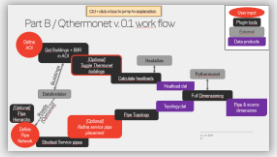
To perform thermonet dimensioning open the **Full_dimensioning** tool

- Input the **Heatloads.dat** file created in part B1, and the **Topology.dat** file created in Part B2.
- Select either Borehole Heat Exchanger (BHE) or Horizontal Heat Exchanger (HHE) source mode.

Note: The output file is not currently used – output is written to the Python Console in QGIS.





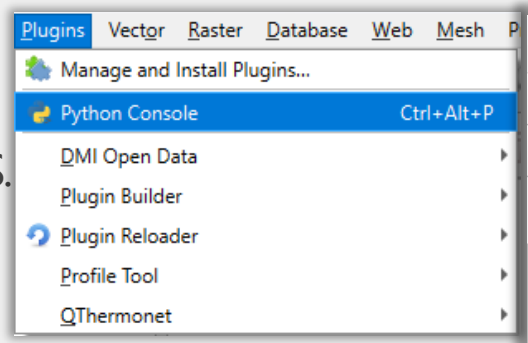


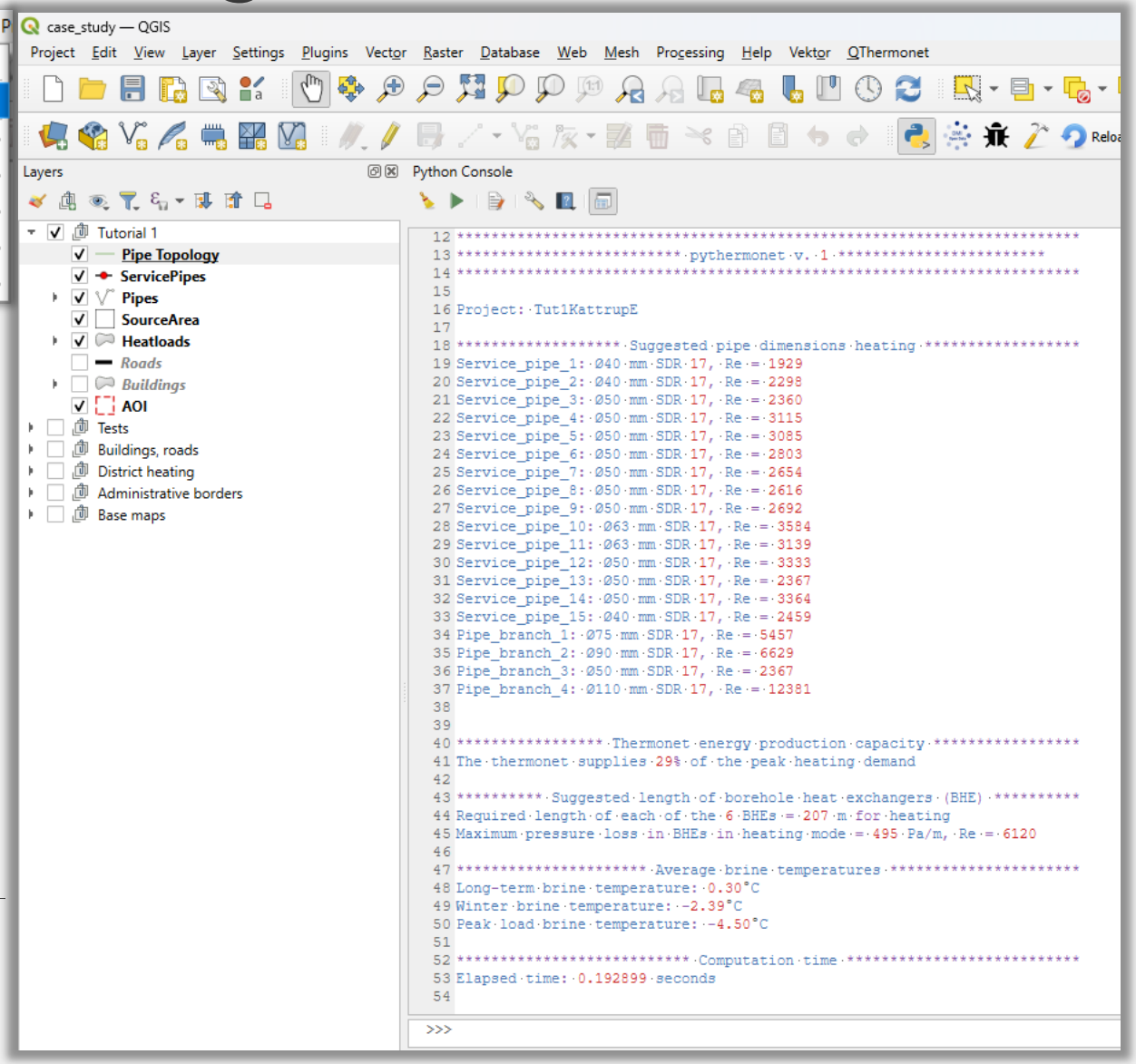
Part B3 / Full dimensioning 2

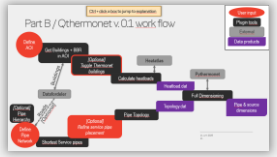
After running the tool, the output is written to the Python Console in QGIS.

The output contains:

1. The suggested pipe dimensions for heating.
2. The percent of total peak demand covered by the pipe network constituting the thermonet (without source area)
3. The dimensions of the thermonet source (BHE/HHE pipe length)
4. Average brine temperatures
5. Computation time







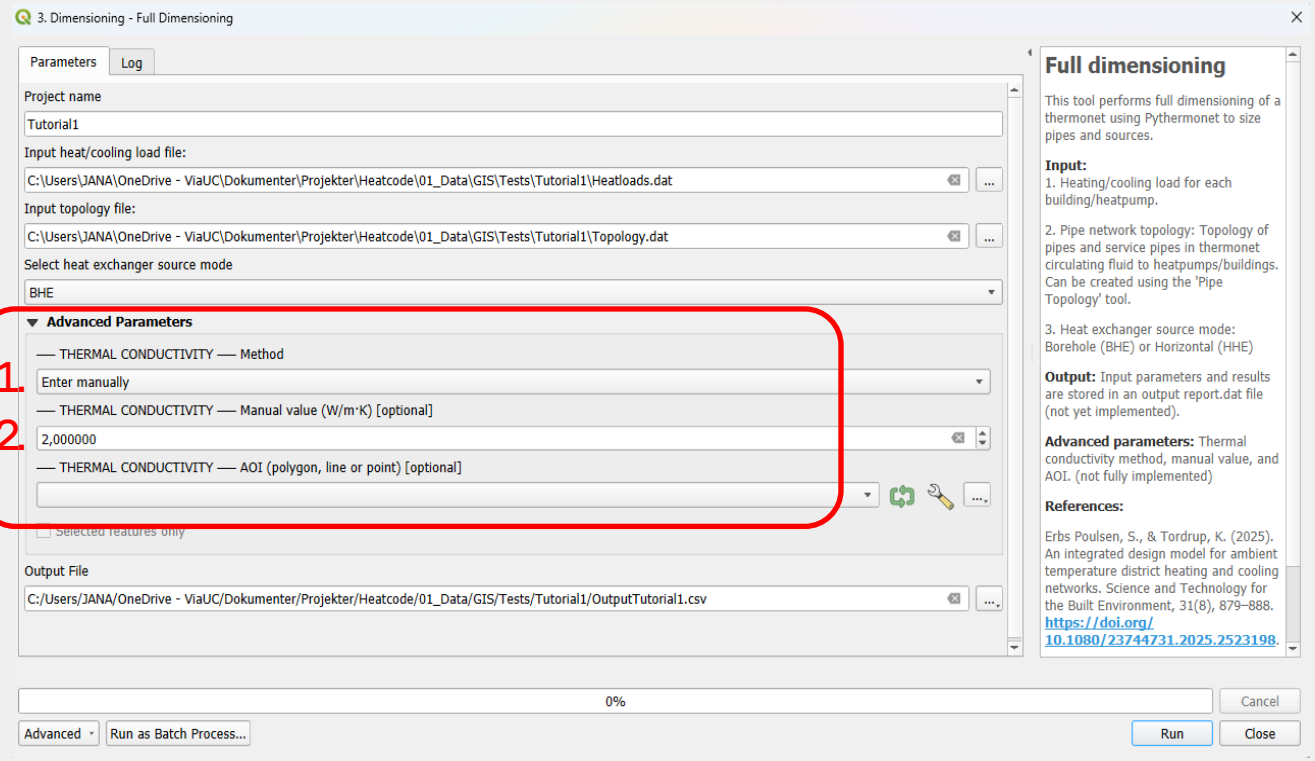
Part B3 / Ground thermal conductivity

In the advanced parameter section you can :

1. Input a ground thermal conductivity manually (default is 2 W mK^{-1}),

OR

2. Calculate the thermal conductivity based on a GEUS API call using the method described above (no report is generated). An AOI (polygon, line, or point with a single element) is required to locate your study area. This method only works in Denmark.



Plugin development plans

- Write output to a file - including the parameters + assumptions used in the thermonet dimensioning.
- Tailor (separate) json output to Modelica…
- Enable more sophisticated handling of winter/daily and total heat load calculations (EnergyPlus)
- Easy adjustment of building parameters for individual buildings
- Enable multiple heat sources (e.g. waste heat / LEG-DHC), seasonal demand/supply, balancing thermonet
- Enable different configurations (Now: tree structure, single borefield)
- Link to GHEDesigner for automatic /optimized borehole locations within borefield?
- Educational/display of existing thermonets
- …

Bring ideas to life
VIA University College

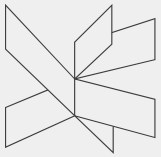
If you need any assistance, please contact:

kri@via.dk

Developers: Jane Lund Andersen, Kristoffer Bested Nielsen, Matthias Larsen,

2024-2026

Bring ideas to life
VIA University College



QThermonet

A thermonet dimensioning tool for QGIS